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DI PAVIA

Dipartimento di Fisica
Corso di laurea in Fisica

**Muon Lifetime Experiment:
analysis of the muon behaviour inside
a hodoscope**

Thesis relation for the degree of:
Cecilia Bombari
Matricola 503808

Supervisor:
Prof.ssa Daniela Rebuzzi
Cosupervisor:
Prof. Oleg Brandt

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Physics is a parameterization of our ignorance

Abstract (English)

This thesis presents an in-depth investigation of muon behaviour within a cubic array of 25 polyvinyltoluene (PVT) plastic scintillators, conducted during a summer internship at the Cavendish Laboratory in Cambridge. After a shrewd theoretical introduction to the muon decay and behaviour through matter, the practical experience in the laboratory is explored in detail.

Initially, the work focused on developing Python software for bit-by-bit decoding of scintillation data. The data was recorded by scintillators paired with photomultiplication sensors, which were configured inside the scintillator structure to detect muon events. This initial step enabled the decoding and analyzing the number of hits per scintillator within the 5x5 array.

A novel automated threshold scan was performed to assess the efficiency of the silicon photomultipliers (SiPMs). This automated process, implemented in the software, drastically reduced the time required for the threshold scan compared to the previous manual procedure.

The hardware setup was also a major focus. The assembly of the experimental apparatus, including new components and printed circuit boards, was followed by detailed data analysis with the custom software. A significant portion of the work involved debugging both the firmware used for data acquisition and resolving hardware-related issues, as well as investigating possible statistical and systematic errors in both the electronics and bit analysis.

My GitHub repository:

<https://github.com/ceciliabombari/MuonHodoscope>

The Cambridge HEP Muon Hodoscope page:

<https://www.hep.phy.cam.ac.uk/hodoscope>

Abstract (Italiano)

Questa tesi riguarda una indagine approfondita del comportamento del muone all'interno di un array di 25 scintillatori plastici in poliviniltoluene (PVT), condotta durante un periodo di sei settimane presso il Cavendish Laboratory di Cambridge. Dopo un'introduzione teorica sul decadimento del muone, sul suo comportamento attraverso la materia e sui sistemi di rivelazione, l'esperienza laboratoriale è stata esaminata in dettaglio.

Il lavoro iniziale ha riguardato lo sviluppo di un software in Python per la decodifica bit a bit dei dati provenienti dall'apparato di scintillazione. I dati sono stati registrati dagli scintillatori accoppiati a sensori fotomoltiplicatori, configurati all'interno della struttura dello scintillatore per rilevare eventi muonici.

Una scansione di soglia automatizzata, completamente nuova, è stata implementata per valutare l'efficienza dei fotomoltiplicatori al silicio (SiPM). Questo processo ha ridotto drasticamente il tempo necessario per la scansione rispetto alla procedura manuale utilizzata in precedenza.

Il lavoro ha coinvolto anche l'assemblaggio dell'apparato sperimentale, inclusi nuove componenti e circuiti stampati, seguito da un'analisi approfondita dei dati con il software sviluppato. Una parte significativa del lavoro è stata dedicata al debugging del firmware utilizzato per l'acquisizione dei dati e alla risoluzione di problemi relativi all'hardware, così come all'analisi dettagliata delle fonti di errore del sistema elettronico e dell'acquisizione dati.

La mia repository GitHub:

<https://github.com/ceciliabombari/MuonHodoscope>

Sito di riferimento Cambridge HEP:

<https://www.hep.phy.cam.ac.uk/hodoscope>

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Chapter 1

Theoretical Background

A **muon** (μ) is a fundamental particle that closely resembles the electron, possessing an electric charge of $-1\ e$ and spin $\frac{1}{2}$, but with a significantly higher rest mass, which is $105.66\ \text{MeV}/c^2$, approximately 207 times the mass of the electron. It is categorized as a second-generation lepton inside the Standard Model. This peculiar particle presents also an anomalous magnetic moment, which has long been the subject of state-of-the-art research, including the $g - 2$ experiments at Fermilab [3]

. The muon is also a transient subatomic particle with an average lifetime of $2.2\ \mu\text{s}$, much longer than many other subatomic entities. When generated by secondary cosmic rays¹, it can be easily detected on Earth thanks to its relativistic speed and the consequent time dilation phenomenon. Like the decay of a free neutron (with a lifetime of around 15 minutes), muon decay is relatively slow as driven only by the weak force [11][9].

Who ordered that?

The discovery of the muon was one of the most unexpected events in the history of particle physics. In 1936, Carl Anderson and his student Seth Neddermeyer lurched upon an unusual signal while studying cosmic rays at Caltech. They were investigating the trajectory of particles produced by cosmic radiation using a cloud chamber² and a magnetic field. The particles they observed seemed to be heavier than electrons, yet lighter than

¹Secondary cosmic rays refer to the particles produced when primary cosmic rays from space collide with atoms in the Earth atmosphere. These interactions result in cascades of particles, including muons, which can be detected at the Earth surface.

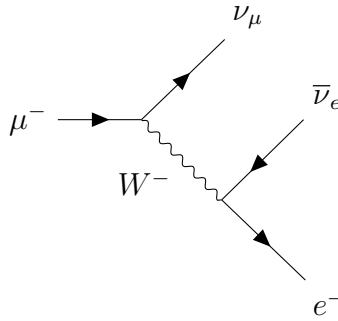
²A cloud chamber, often referred to as a *Wilson cloud chamber*, is a device used to detect and visually track the movement of ionizing radiation. It works by creating a sealed environment filled with a supersaturated vapor, typically made of water or alcohol, allowing the paths of charged particles to become visible.

protons. However, they did not fit into the known framework of particles at the time. This unusual particle didn't behave like a proton or neutron, nor did it exhibit the same characteristics as the electron. They initially called it *mesotron*, because its behaviour *stood in the middle*, as mesons did.

After further study, they concluded that they had discovered a new type of particle, which came to be known as the muon. This discovery was puzzling to physicists because the muon appeared to have no clear role in the existing theories of matter, especially since it was not involved in strong force processes, like mesons. As the physicist I.I. Rabi famously remarked, *Who ordered that?* – highlighting the confusion surrounding this discovery. [7].

1.1 The muon decay

The muon decay almost always results in at least three particles, which include an electron with the same charge as the muon and two varieties of neutrinos, in order to preserve both charge and the leptonic number through the entire process. Since it happens solely via this interaction, Muon decay is considered a fundamental staple in probing the weak interaction, defining its role in subatomic processes. The weak force is, in fact, responsible for phenomena such as beta decay in nuclear structures.



The decay of a muon (μ^-) occurs via the following reactions:

$$\mu^- \rightarrow e^- + \bar{\nu}_e + \nu_\mu$$

Where ν_e and ν_μ are, respectively, the electron neutrino and the muon neutrino, and $\bar{\nu}_e$ and $\bar{\nu}_\mu$ are their corresponding antineutrinos. The mean lifetime of a free muon is:

$$\tau = 2.197 \mu s$$

The decay of a particle follows a Poisson process, which is a type of Markovian process³. Given a muon at rest, generated at time $t = 0$, the probability that it will survive until time t is described by an exponential distribution:

$$P(t) = e^{-t/\tau}$$

Given N_0 muons at time $t = 0$, the number of muons remaining at any time t is then:

$$N(t) = N_0 e^{-t/\tau}$$

1.1.1 Fundamental Forces

In physics, the fundamental interactions or fundamental forces are the interactions or forces of nature that allow the description of physical phenomena at all distance and energy scales, and thus cannot be reduced to other forces. Four fundamental forces or interactions have been identified: gravitational interaction, electromagnetic interaction, weak interaction, and strong interaction [1].

1.1.2 The weak charged-current interaction

Quantum electrodynamics (QED) and quantum chromodynamics (QCD) are the quantum field theories describing the electromagnetic and strong forces, respectively. Apart from some commutation algebraic differences, these theories surprisingly share various common aspects, such as mediation by a massless neutral spin-1 boson (photons and gluons) and the mathematical form of the vertices of interaction and matrix element through which they are described according to Feynman rules.

The weak interaction, instead, seems not to follow this pattern. First of all, its mediator is a massive, charged boson, $W^{\pm 4}$ [19].

Another point to be highlighted is parity conservation. It can be proven that, for both QED and QCD vertices, parity is conserved, whereas this is not the case for the weak interaction, as demonstrated by Madame Wu's experiment. The form of the four-current must be changed, taking on a different and non-trivial form.

Parity is an operation that coincides with spatial coordinates inversion,

³Stochastic process that exhibits the property of "memorylessness." In this context, the future state of the system depends only on its present state and not on the sequence of events that preceded it.

⁴It can also be described by a neutral massive boson, Z , but in this analysis of charged currents responsible for the muon decay, this will not be explored further.

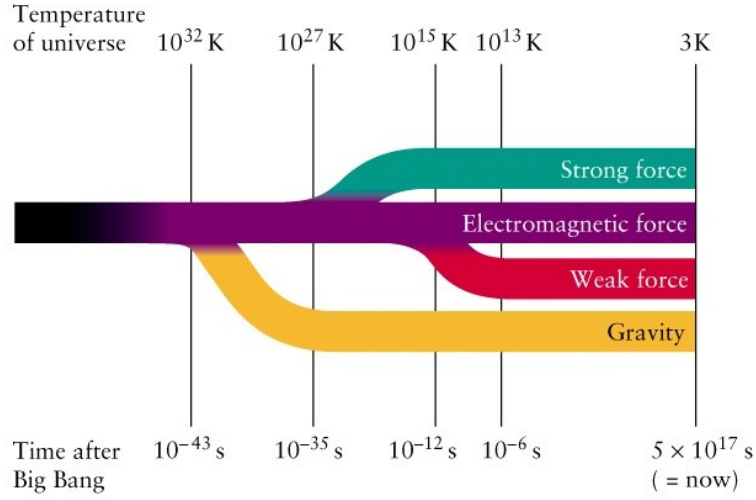


Figure 1.1: Division of the fundamental forces from the initial singularity after the Planck time (10^{-43} s). The forces bifurcated as the Universe cooled: first gravity, followed by the strong force, and finally the weak and electromagnetic forces, which remained unified in the electroweak force until approximately 10^{-12} seconds after the Big Bang. Source: Wikimedia Commons.

that is $\mathbf{x} \rightarrow -\mathbf{x}$. It could be promoted to a quantum operator, defining the transformation of the quantum state

$$\Psi(\mathbf{x}, t) \rightarrow \Psi'(\mathbf{x}, t) = \hat{P}\Psi(\mathbf{x}, t) = \Psi(-\mathbf{x}, t)$$

It comes naturally that applying it twice, the original wavefunction returns. As a matter of fact, this operator is its inverse itself,

$$\hat{P}^2 = I$$

Additionally, if physics is invariant under this operation, parity is a well-defined unitary and Hermitian operator. Whenever it commutes with the Hamiltonian, it is assessed to be an observable *conserved* quantity in the interaction, as well as the state is also an eigenstate of the parity transformation. It is thus a trivial result that parity eigenvalues correspond to ± 1 .

Let us briefly demonstrate the case for electromagnetic and strong force theories. First of all, the *matrix element* is a crucial quantity used to describe the probability amplitude of a particle interaction or scattering process. It represents the transition - *braket* - between the initial and final quantum states under the influence of an interaction Lagrangian density $\mathcal{L}_{\text{int}}$. The

matrix element is thus $\mathcal{M} = \langle f | \mathcal{H}_{\text{int}} | i \rangle$, where the Hamiltonian is derived from the Lagrangian density. The relation between the Hamiltonian and the Lagrangian density is given by:

$$\mathcal{H} = \sum_i \pi_i \dot{\phi}_i - \mathcal{L},$$

where π_i are the canonical momenta conjugate to the fields ϕ_i , and $\dot{\phi}_i$ are their time derivatives. This formulation translates the energy density from the Lagrangian to the Hamiltonian formalism. Its square module, $|\mathcal{M}|^2$, is related to physical observables like cross-sections, probability amplitudes, and decay rates.

In quantum electrodynamics (QED) and quantum chromodynamics (QCD), parity conservation can be analyzed by examining the matrix elements describing particle interactions. In both theories, parity is preserved, although the mechanisms differ slightly due to the additional structure present in QCD, primarily involving color factors [19].

In QED, the electromagnetic interaction between charged particles is mediated by the photon. The interaction *Lagrangian density* can be written as:

$$\mathcal{L}_{\text{int}} = e \bar{\psi} \gamma^\mu A_\mu \psi,$$

where e is the electric charge, ψ is the Dirac field for the fermion (e.g., the electron), and A_μ is the photon field. Let us consider the scattering of two electrons, $e^- + e^- \rightarrow e^- + e^-$ (i.e., Møller scattering). The matrix element for this process can be expressed as:

$$\mathcal{M} \sim \bar{u}(p') \gamma^\mu u(p) \frac{g_{\mu\nu}}{q^2} \bar{u}(k') \gamma^\nu u(k),$$

where p and k are the initial momenta of the electrons, p' and k' are the final momenta, and q is the momentum transfer. Under a parity transformation, the vector four-current $j^\mu = \bar{u}(p') \gamma^\mu u(p)$ changes sign (in particular, the space-like component of j^μ), but this transformation is canceled by the four-vector scalar product $j^\mu j_\mu$, ensuring that the overall interaction remains invariant. Thus, parity is conserved in QED.

In QCD, the situation is more complex due to the involvement of color charge and the gluon field. The interaction *Lagrangian density* in QCD is:

$$\mathcal{L}_{\text{int}} = g_s \bar{\psi} \gamma^\mu T^a G_\mu^a \psi,$$

where g_s is the strong coupling constant, T^a are the generators of the SU(3) color group, and G_μ^a represents the gluon field. The matrix element for quark-quark scattering (e.g., $q + q \rightarrow q + q$) takes the form:

$$\mathcal{M} \sim \bar{u}(p')\gamma^\mu T^a u(p) \frac{g_{\mu\nu}}{q^2} \bar{u}(k')\gamma^\nu T^a u(k),$$

similar to QED, but with the addition of color factors T^a . The quark current, $\bar{u}(p')\gamma^\mu T^a u(p)$, behaves under parity in the same way as in QED. Since the parity transformation affects both the quark spinor and the gluon field consistently, parity is conserved in QCD as well, apart from color factors, which remain unaffected by spatial inversion.

Parity violation

When it comes to the weak interaction, some problems crop up. It was in 1957 when Wu and collaborators leached into nuclear β -decay of polarised cobalt-60. The cobalt decays by transitioning to an excited state of nickel:



A neutron in the cobalt nucleus transforms into a proton ($Z \rightarrow Z + 1$), emitting an electron and an antineutrino.

The excited nickel then decays into the stable form of nickel by emitting two gamma rays:



In the presence of these photons, the nickel excited state is known to be subject to electromagnetic interactions, which conserve parity. Therefore, the distribution of the two photons in the final state should be isotropic (i.e., this distribution should be symmetric with respect to the magnetic moment of the cobalt that should be aligned with the field).

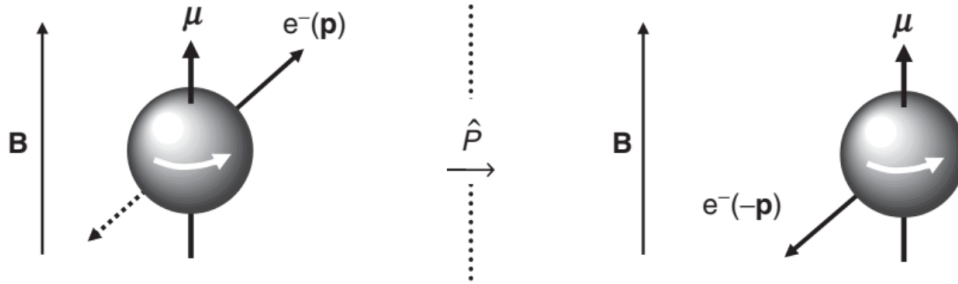


Figure 1.2: The β -decay of polarised ${}^{60}\text{Co}$. On the left, an electron is emitted in a particular direction. On the right, the parity inverted equivalent is shown [19].

The observation, hypothesized by Lee and Yang, is the following: The magnetic moment μ and the magnetic field $\mathbf{B}$ are pseudo-vectors and thus remain invariant under parity transformation. The only thing that changes is the momentum of the emitted electron, which is a vector.

If parity is conserved, under the operator $\hat{P}$, the electron emissions in arbitrary directions $+\mathbf{p}$ and $-\mathbf{p}$ should occur with equal probability. In this case, under parity transformation, the action of $\hat{P}$ would generate exactly

the same decay pattern.

However, the experiment found that the action of $\hat{P}$ does not preserve the symmetry: there is a clear asymmetry in the emission of the electron in the decay process, with a preference for emission in the direction opposite to the cobalt/nickel magnetic moment.

This leads to the conclusion that *parity is NOT conserved in weak interactions*. Madame Wu experiment is of paramount importance in particle physics panorama: not only does it state the maximal violation of parity, but also confirms the Goldhaber superselection⁵. Like the contributions of many women in science at that time, Dr. Chien-Shiung Wu's work was not acknowledged and the Nobel Prize was indeed awarded to her collaborators Lee and Yang in 1957.

V-A structure of the weak interaction

Having stated the violation of parity, the weak interaction vertex needs to be of a different, nontrivial form with respect to the one of QED and QCD theories. On top of this, the interaction should remain Lorentz-invariant⁶. It can be demonstrated that the most general Lorentz-invariant form for the interaction between a fermion and a boson is a linear combination of the bilinear covariants⁷. When restricted to the exchange of a spin-1 (vector)

Type	Form	Components	Boson spin
Scalar	$\bar{\psi}\phi$	1	0
Pseudoscalar	$\bar{\psi}\gamma^5\phi$	1	0
Vector	$\bar{\psi}\gamma^\mu\phi$	4	1
Axial vector	$\bar{\psi}\gamma^\mu\gamma^5\phi$	4	1
Tensor	$\bar{\psi}(\gamma^\mu\gamma^\nu - \gamma^\nu\gamma^\mu)\phi$	6	2

Table 1.1: Lorentz-invariant bilinear covariant currents.

boson, the interaction takes the following form:

$$j^\mu \propto \bar{u}(p)(g_V\gamma^\mu + g_A\gamma^\mu\gamma^5)u(p) = g_V j_V^\mu + g_A j_A^\mu,$$

⁵Experiment carried out by Goldhaber in 1958 in which is stated that, for a principle of nature, particles are left-handed, while antiparticles are right-handed.

⁶the final probability amplitude for the process remains unchanged under Lorentz transformations (changes in the observer inertial frame). This is a fundamental requirement for any relativistic quantum field theory, as it ensures that physical predictions are consistent with the theory of special relativity.

⁷combinations of individual γ -matrices that can be combined into a Lorentz-invariant matrix element.

where g_V and g_A are the vector and axial vector coupling constants, respectively. The current is decomposed into vector and axial vector components:

$$j_V^\mu = \bar{u}(p)\gamma^\mu u(p), \quad j_A^\mu = \bar{u}(p)\gamma^\mu\gamma^5 u(p).$$

What happens under parity transformation? The result on the purely vectorial current is already been faced in 1.1.2. As far as the axial vector component is concerned, it can be demonstrated that the time-like component of the current does change sign under parity, while the space-like does not. This implies that the scalar product of two axial vector currents is invariant under parity transformations.

$$j_1 \cdot j_2 = j_1^0 j_2^0 - j_1^k j_2^k \xrightarrow{\hat{P}} (-j_1^0)(-j_2^0) - j_1^k j_2^k = j_1 \cdot j_2.$$

This result is expected, as the matrix element represents a scalar quantity. If it is constructed from the scalar product of two four-vectors — whether they are vectors or axial vectors — it must remain invariant under a parity transformation.

To summarize, the behaviour of the vector and axial vector current components under parity transformations is given by:

$$\begin{aligned} j_V^0 &\xrightarrow{\hat{P}} +j_V^0, & j_V^k &\xrightarrow{\hat{P}} -j_V^k, \\ j_A^0 &\xrightarrow{\hat{P}} -j_A^0, & j_A^k &\xrightarrow{\hat{P}} +j_A^k. \end{aligned}$$

While the scalar products of two vector currents or two axial vector currents remain unchanged under a parity transformation, the scalar product $j_V \cdot j_A$ transforms as $-j_V \cdot j_A$. *This combination of vector and axial vector currents plays a crucial role in explaining the parity violation observed in weak interactions.* It can be demonstrated that, in the case both g_V and g_A are equal to zero, parity is conserved in the interaction, whereas it is maximally violated whenever $|g_V| = |g_A|$, so with a $V + A$ or $V - A$ structure. $V - A$ (*vector minus axial*) is the right form for the weak charged current, experimentally proved for the exchange of a $W^\pm$ boson, indeed. Hence the vertex of weak interaction is written as:

$$-\frac{ig_W}{\sqrt{2}} \frac{1}{2} \gamma^\mu (1 - \gamma^5), \tag{1.1}$$

where g_W corresponds the weak coupling constant (sometimes simply g), while the numerical factors are a matter of normalisation. The four-vector current is thus written as

$$j^\mu = \frac{g_W}{\sqrt{2}} \bar{u}(p') \frac{1}{2} \gamma^\mu (1 - \gamma^5) u(p).$$

Chirality in the weak current

Introducing the chiral projectors

$$P_R = \frac{1}{2}(1 + \gamma^5) \quad P_L = \frac{1}{2}(1 - \gamma^5)$$

any spinor can be decomposed in this way

$$u = \frac{1}{2}(1 + \gamma^5)u + \frac{1}{2}(1 - \gamma^5)u = P_R u + P_L u = a_R u_R + a_L u_L,$$

In the weak interaction, the vertex for the $V - A$ form from equation 1.1 already contains the left-handed chiral projection operator:

$$\frac{1}{2}(1 - \gamma^5).$$

Thus, the current with two right-handed spinors is:

$$j_{RR}^\mu = \frac{g_W}{\sqrt{2}} \bar{u}_R(p') \gamma^\mu \frac{1}{2}(1 - \gamma^5) u_R(p) = \frac{g_W}{\sqrt{2}} \bar{u}_R(p') \gamma^\mu P_L u_R(p) = 0,$$

since the projection operator P_L annihilates u_R . Hence, the only non-zero current for particle spinors involves left-handed chiral states, meaning that *only left-handed particle states contribute to the charged weak interaction*. For antiparticle spinors, P_L projects out right-handed chiral states:

$$\frac{1}{2}(1 - \gamma^5)v = v_R,$$

implying that only right-handed chiral antiparticle states participate in the charged-current weak interaction. In the limit $E \gg m$, where chirality and helicity coincide, the $V - A$ vertex selects left-handed helicity⁸ particle states and right-handed helicity antiparticle states, allowing only certain helicity configurations for weak interaction vertices involving leptons and neutrinos. This maximal coupling of the weak charged current to left-handed chiral states in particles, and right-handed chiral states in antiparticles, explains once more parity violation in weak interactions.

⁸In particle physics, helicity is defined as the projection of a particle's spin onto the direction of its momentum. For massless particles, helicity corresponds to chirality, whereas for massive particles, they are distinct concepts.

1.1.3 The Fermi theory

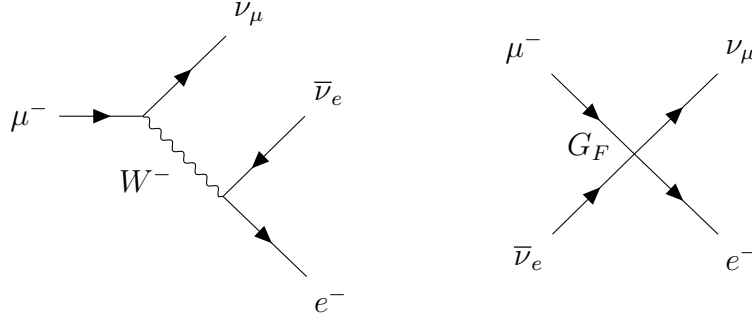
Let us consider primarily, as a consequence of Feynman rules, the mathematical form of the weak interaction propagator, which is, considering the massive nature of the vectorial boson W ($m_W \sim 80 \text{ GeV}$):

$$\frac{-ig_{\mu\nu}}{q^2 - m_W^2} \quad (1.2)$$

yet dependence from the four-momentum can be removed in the case of weak interactions (i.e. $|g^2| \ll m_W^2$)

$$\frac{ig_{\mu\nu}}{m_W^2} \quad (1.3)$$

Physically speaking, this approximation corresponds to a contact interaction. In the specific case of the muon decay, it corresponds to a four-fermion contact interaction approximation, commonly reported as the *Fermi contact interaction*, a non renormalizable theory⁹ which becomes instead extremely efficient for calculation and order of magnitude for this type of interaction at low energy scales.



Focusing on the Fermi Feynman diagram on the left, the matrix element for the muon decay can be simply written in terms of a contact interaction

$$\mathcal{M}_{\text{Fermi}} \propto G_F [\bar{u}(p_e)\gamma^\mu(1 - \gamma^5)u(p_{\nu_e})] [\bar{u}(p_{\nu_\mu})\gamma_\mu(1 - \gamma^5)u(p_\mu)], \quad (1.4)$$

where:

- G_F is the Fermi constant,

⁹A theory is said to be *renormalizable* if, in the process of calculating quantum corrections, it is possible to absorb the divergences into the parameters of the theory, such as coupling constants and masses, through a finite number of counterterms. In this way, the theory remains predictive at all energy scales.

- $u(p)$ are spinors for the involved particles,
- $\gamma^\mu(1 - \gamma^5)$ represents the $V - A$ structure to justify parity violation.

Considering parallelly the Standard Model description, the weak interaction is mediated by the W^- boson, and the matrix element for the same process (e.g., muon decay) with the W^- boson propagator is:

$$\mathcal{M}_{W\text{-boson}} \propto \frac{g_W^2}{q^2 - m_W^2} [\bar{u}(p_e)\gamma^\mu(1 - \gamma^5)u(p_{\nu_e})] [\bar{u}(p_{\nu_\mu})\gamma_\mu(1 - \gamma^5)u(p_\mu)],$$

where:

- g_W is the weak coupling constant,
- q^2 is the momentum transfer in the interaction,
- m_W is the mass of the W^- boson.

In the low-energy limit, where $q^2 \ll m_W^2$, the propagator can be written as 1.3. The matrix element for the propagator interaction can be approximated as:

$$\mathcal{M}_{W\text{-boson}} \propto \frac{g_W^2}{m_W^2} [\bar{u}(p_e)\gamma^\mu(1 - \gamma^5)u(p_{\nu_e})] [\bar{u}(p_{\nu_\mu})\gamma_\mu(1 - \gamma^5)u(p_\mu)]. \quad (1.5)$$

By comparing equation 1.5 with 1.4, we identify the relation between the Fermi constant G_F and the weak coupling constant g_W ¹⁰:

$$G_F = \frac{\sqrt{2}g_W^2}{8m_W^2} \quad (1.6)$$

Thus, the Fermi constant G_F is related to the weak coupling constant g_W and the mass of the W^- boson m_W , providing a consistent description of weak interactions both in the low-energy Fermi theory and the full electroweak theory with the W^- boson propagator. The strength of the weak interaction is best determined from low-energy experiments, particularly through the measurement of the muon lifetime. The decay rate for the process $\mu^- \rightarrow e^- \bar{\nu}_e \nu_\mu$ requires integrating over the three-body phase space of the final states, which is not further analyzed in this context. The resulting expression relates the muon lifetime τ_μ to its mass as follows:

¹⁰up to the constant $\frac{\sqrt{2}}{8}$ due to consistencies taken into account for normalization and conventions.

$$\Gamma(\mu^- \rightarrow e^- \nu_\mu \bar{\nu}_e) = \frac{1}{\tau_\mu} = \frac{G_F^2 m_\mu^5}{192\pi^3}. \quad (1.7)$$

Precise measurements of the muon mass and lifetime,

$$m_\mu = 0.1056583715(35) \text{ GeV} \quad \text{and} \quad \tau_\mu = 2.1969811(22) \times 10^{-6} \text{ s},$$

allow for a very accurate determination of the Fermi constant:

$$G_F = 1.16638 \times 10^{-5} \text{ GeV}^{-2}.$$

However, G_F does not directly represent the fundamental strength of the weak interaction. The true coupling strength is tied to the ratio between the weak coupling constant g_W and the mass of the W -boson, as discussed earlier in equation 1.4. Nevertheless, G_F remains the most precisely measured parameter from muon decay and is used to express the effective strength of the weak interaction at low energies.

The fundamental weak coupling constant g_W can be deduced from G_F using the precise determination of the W -boson mass, $m_W = 80.377 \pm 0.011 \text{ GeV}$ [10]. From this, the relation between G_F and g_W is established and the dimensionless coupling constant can be calculated:

$$\alpha_W = \frac{g_W^2}{4\pi} = \frac{8m_W^2 G_F}{4\sqrt{2}\pi} \approx \frac{1}{30}.$$

This indicates that the weak force is inherently stronger than the electromagnetic force, as $\alpha_W > \alpha \approx \frac{1}{137}$. The apparent weakness of the weak interaction at low energies is primarily due to the large mass of the W boson, which appears in the propagator. The mass of the W boson suppresses the strength of the interaction at low energy, making it seem weaker than quantum electrodynamics.

1.1.4 Lepton universality

Not only do experiments adhere to theory, but even show some magic behind the muon decay. In fact, the weak interaction has the same strength across all lepton flavours. For instance, the muon decay $\mu^- \rightarrow e^- \bar{\nu}_e \nu_\mu$ involves two vertices of weak interaction: $\mu^- \nu_\mu W$ and $W e^- \bar{\nu}_e$. In theory, the weak couplings at these two vertices could differ, so in general the decay rate for muon decay from equation 1.7 is rewritten as:

$$\Gamma(\mu^- \rightarrow e^- \bar{\nu}_e \nu_\mu) = \frac{1}{\tau_\mu} = \frac{G_F^{(e)} G_F^{(\mu)} m_\mu^5}{192\pi^3}, \quad (1.8)$$

Performing the same calculation for the decay of the tau lepton $\tau^- \rightarrow e^- \bar{\nu}_e \nu_\tau$ results in:

$$\Gamma(\tau^- \rightarrow e^- \bar{\nu}_e \nu_\tau) = \frac{G_F^{(e)} G_F^{(\tau)} m_\tau^5}{192\pi^3}. \quad (1.9)$$

1.8 and 1.9 are equal, except for the constant G_F , which refers to the τ , and another G_F which refers to the μ or e .

Discussing lepton universality means experimentally verifying that these couplings, namely the constants G_F , which remind us of the existence of an effective theory at low energy, are independent of the lepton number of the lepton that decays. This is not an assumption in the model but must be verified through experimental results. Experimentally, these two decay channels are well-separated; in fact, detectors distinguish between an e^- and a μ^- with high precision. Performing the experimental measurements, we obtain:

$$\frac{G_F^{(e)}}{G_F^{(\mu)}} = 1.0023 \pm 0.0033$$

$$\frac{G_F^{(\tau)}}{G_F^{(\mu)}} = 1.000 \pm 0.004$$

This means that, within experimental errors, the Fermi couplings G_F are the same for each lepton family:

$$G_F^{(e)} = G_F^{(\mu)} = G_F^{(\tau)} = G_F$$

When the strength of the weak interaction is the same for every lepton flavour, the process containing a μ carries the same weight as that containing a τ in terms of the weak interaction coupling.

1.2 The muon through matter

Particles such as electrons, protons, muons and neutrinos interact with matter primarily through electromagnetic interactions, ionization, and scattering. Muons, being relatively heavy charged particles, typically interact with matter via ionization, as described by the Bethe-Bloch equation.

1.2.1 Bethe-Bloch equation

Firstly we define

$$W_{\max} = \frac{2m_e c^2 \beta^2 \gamma^2}{1 + 2\gamma \frac{m_e}{M} + \left(\frac{m_e}{M}\right)^2}$$

as the maximum possible energy transfer to an electron in a single collision (expressed in MeV) and I the average energy required to excite or ionize an atom in the material¹¹. At intermediate energies (i.e. typical energies muons arrive on the Earth surface with $\beta\gamma$ from 0.1 to 1000), the energy loss per unit length for cosmic-rays muons crossing a specific material (the absorber), being moderately relativistic charged particles, is well described by the Bethe-Bloch equation:

$$\left\langle -\frac{dE}{dx} \right\rangle = K z^2 \frac{Z}{A} \frac{1}{\beta^2} \left[\frac{1}{2} \ln \left(\frac{2m_e c^2 \beta^2 \gamma^2 W_{\max}}{I^2} \right) - \beta^2 - \frac{\delta(\beta\gamma)}{2} \right] \quad (1.10)$$

The formula provides an accurate approximation of energy loss within a range of $\beta\gamma$ from 0.1 to 1000, with an accuracy within a few percent. At $\beta\gamma \approx 0.1$, energy losses become comparable to atomic electron velocities, and at $\beta\gamma \approx 1000$, radiative losses start to become significant. All the quantities are specified in table 1.2.

Both high and low-energy limits depend on Z , and there is a minor correction to the formula due to the projectile particle mass, M , although for practical purposes this correction is small, and the stopping power, also reported as *mass stopping power*, is primarily a function of $\beta\gamma$ [8].

The stopping power initially decreases as $\frac{1}{\beta^\alpha}$, where $\alpha \approx 1.7$ – 1.5 , and continues to decrease as the atomic number Z increases. It reaches a broad minimum at $\beta\gamma = 3.8$ – 3.0 as Z rises from 6 to 82, after which it gradually increases due to the growth in the argument of the logarithmic term. This increase is driven by two primary factors. About two-thirds of the rise

¹¹It is an empirical quantity that depends on the atomic structure of the material and varies with the element being considered, and ranges from 10 eV to 1000 eV.

Symbol	Description	Unit of Measurement
K	Coefficient for $\langle \frac{dE}{dx} \rangle$	$\text{MeV} \cdot \text{mol}^{-1} \cdot \text{cm}^2$
z	Charge number of the incident particle	—
Z	Atomic number of the absorber	—
A	Atomic mass of the absorber	$\text{g} \cdot \text{mol}^{-1}$
β	Velocity of the particle relative to c	—
γ	Lorentz factor	—
I	Mean excitation energy	eV
W_{max}	Maximum energy transfer	MeV
M	mass of the particle	MeV/c^2

Table 1.2: Quantities present in the Bethe-Bloch formula.

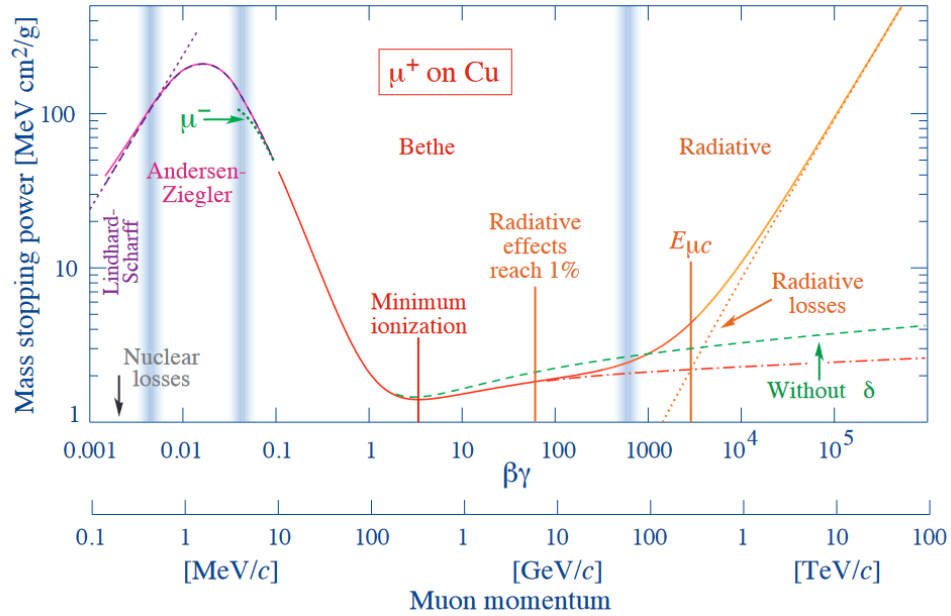


Figure 1.3: The mass stopping power (dE/dx) for positive muons in copper is shown as a function of $\beta\gamma = \frac{p}{mc}$, covering nine orders of magnitude in momentum (and twelve orders of magnitude in kinetic energy). The short dotted lines labeled “ μ^- ” represent the “Barkas effect”, which illustrates the dependence of stopping power on the projectile charge at very low energies. Image taken from [8].

is due to the explicit $\beta^2\gamma^2$ dependence, which results from the relativistic flattening and extension of the particle's electric field. Instead of producing ionization at increasing distances, the electric field polarizes the medium, canceling the logarithmic increase at high energies. This is accounted for by the *density-effect* correction $\delta(\beta\gamma)$. The remaining third is caused by the $\beta^2\gamma^2$ dependence of $W_{\max}$, the maximum energy transfer to a recoil electron. The mean energy loss behaviour for muons in copper is shown in the *Bethe*

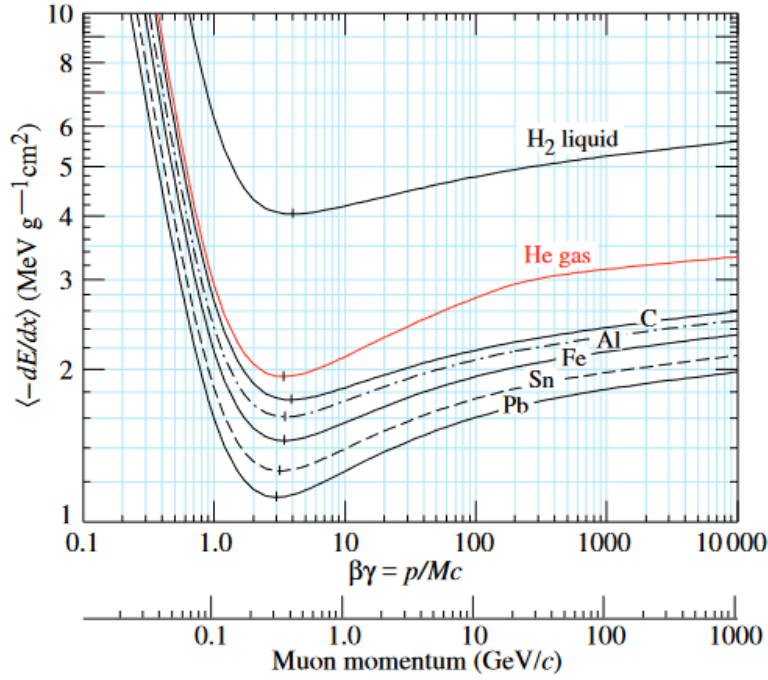


Figure 1.4: Mean energy loss rate in liquid (bubble chamber) hydrogen, gaseous helium, carbon, aluminum, iron, tin, and lead. There are not huge differences among different absorbers: Z/A is proportional to density, yet does not depend strongly on the material[8].

region of Figure 1.3, with radiative effects at higher energies. Only within the *Bethe* region does stopping power depend primarily on β , while mass dependence becomes more complex outside this region. The stopping power for different materials in the *Bethe* region is shown instead in Figure 1.4, where particles with the same velocity have similar energy loss rates across materials, except for hydrogen¹².

¹²A bubble chamber is a device used in particle physics to detect charged particles. It consists of a superheated liquid, often hydrogen, that vaporizes along the path of charged particles, forming tiny bubbles. These bubbles trace the particle trajectory, which can be

As Z increases, there is a slow decrease in the energy loss rate. Differences in behaviour at high energies between gases (e.g., helium) and other materials are due to the density-effect correction $\delta(\beta\gamma)$, described in 1.2.1.

Broad minima in stopping power shift from $\beta\gamma = 3.5$ to 3.0 as Z increases from 7 to 100, with minimum ionization values shown in Figure 1.5. In practical cases, most relativistic particles, such as cosmic-ray muons which have energies below 100 GeV, have energy loss rates near the minimum and are known as *minimum-ionizing particles* (MIPs). They are in fact one of the most penetrating particle species and they usually traverse the entire particle detector, leaving a trail of ionization. This characteristic can be widely used to identify them as the *most stubborn* particles!

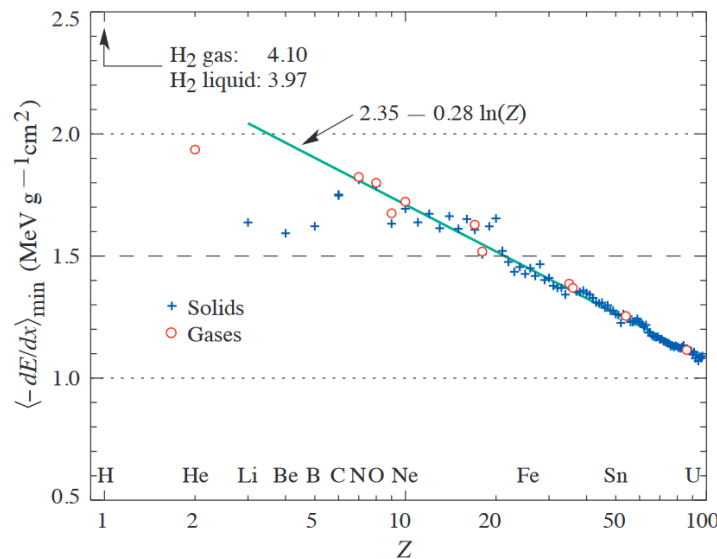


Figure 1.5: The mass stopping power at minimum ionization for chemical elements is shown, with a straight line fit applied for $Z > 6$. However, a linear relationship with atomic number Z is not straightforward, as dE/dx is also influenced by other factors.

Density effect

The density effect serves as a relativistic correction to the simplified Bethe-Bloch formula, originally derived by Bethe in the 1930s. This correction arises due to the significant extension and compression of the electric field produced by a highly relativistic charged particle. The formula that accounts

photographed and analyzed to study interactions such as collisions and decays.

for this correction at high energies is provided in Equation 1.11.

$$\frac{\delta(\beta\gamma)}{2} \rightarrow \ln\left(\frac{\hbar\omega_p}{I}\right) + \ln(\beta\gamma) - \frac{1}{2} \quad (1.11)$$

$$\hbar\omega_p = 28.816 \text{ eV} \sqrt{\frac{Z\rho}{A}} \quad (1.12)$$

Here, $\hbar\omega_p$ denotes the plasma energy of the material, taken from [8]. The material density, ρ , is measured in units of g/cm^3 . The density effect results from the polarization that is induced in the material as the particle passes through it.

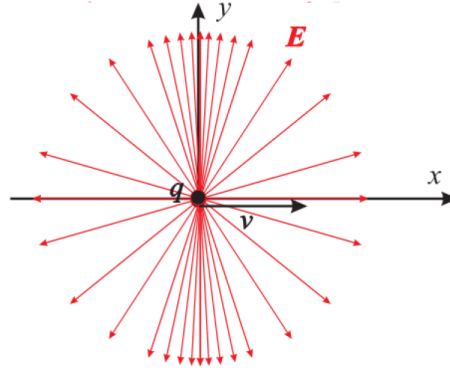


Figure 1.6: Not isotropic distribution of the electric field generated by the relativistic charged particle. Taken from [8].

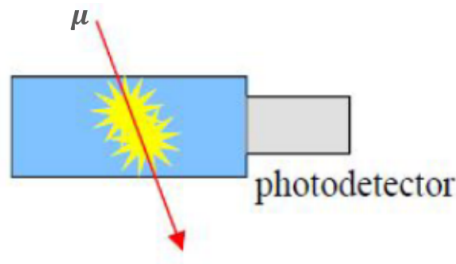


Figure 1.7: Simple sketch of the muon detection, production of light and coupling with the photosensor [13].

1.3 Detecting muons

Muons are, as mentioned previously, highly penetrating particles. A question spontaneously arises: what is the best way to track them?

Organic scintillators are one example. Scintillators, in general, are widely used in modern particle physics experiments that do not require precise spatial resolution. They are an effective solution going hand in hand with relatively low cost.

Liquid and plastic scintillators are two typical prototypes. In particular, detectors based on liquid scintillators have been used in several recent neutrino experiments, such as the Super Kamiokande to look into neutrino oscillation¹³. In an organic scintillator, made of organic carbon molecular chains, the passage of a charged particle leaves some of the molecules in an excited state, whose subsequent relaxation results in the emission of light in the ultraviolet (UV) region. By adding fluorescent dyes, the UV light is absorbed and then re-emitted as visible light in the blue region of the spectrum. This specific re-emission wavelength is known as the **peak emission**, which ensures high compatibility with photomultipliers (SiPMs). The wavelength of this peak emission is in fact carefully selected to match the sensitivity of this optical detectors. SiPMs operate in Geiger mode and detect one photon at a time thanks to photodiodes matrices they're made of, rendering them much more efficient and cheaper than their ancestors PMTs. More details are found in section 2.1.1.

One of the standout features of plastic scintillators is their **fast response time**. When a charged particle, like a muon, passes through a plastic scintil-

¹³Beyond the Standard Model theory stating the non-zero mass of neutrinos, hugely proved by experiments.

lator, the emitted light is produced almost instantaneously, typically within a few nanoseconds. This quick response is crucial in high-energy physics experiments, where precise timing is essential for accurately tracking and differentiating particles. The fast response time of plastic scintillators also allows them to handle high rates of particle detection without significant signal overlap, making them highly suitable for applications involving large particle fluxes, such as cosmic ray or muon detection in scintillator arrays.

Chapter 2

The Muon Lifetime Experiment

In this chapter, I present my contribution to the Muon Lifetime Experiment conducted during August and September 2024 at the Cavendish Laboratory in Cambridge. The project deals with the development of a hodoscope¹, an array of plastic scintillators designed to detect secondary cosmic-ray muons. These cosmic muons can either trespass the setup or come to rest within the array, and their energy selection is based on the Bethe-Bloch formula for PVT (Polyvinyltoluene $(C_9H_{10})_n$), which was defined in Section 1.2.

$$\left\langle -\frac{dE}{dx} \right\rangle = \kappa \frac{Z}{A} \frac{1}{\beta^2} \left(\frac{1}{2} \ln \left(\frac{2m_e \gamma^2 \beta^2 W_{\max}}{I^2} \right) - \beta^2 - \frac{\delta(\beta\gamma)}{2} \right) \rho \quad \text{MeV cm}^{-1} \quad (2.1)$$

Where here PVT density $\rho = 1.023 \text{ g cm}^{-3}$ and mean excitation energy $I = 64.7 \text{ eV}$ [14]. Other key constants in this formula include the atomic mass number $A = 118.170 \text{ g mol}^{-1}$ and atomic number Z , which influence the material response to ionizing particles. The electron mass $m_e = 0.511 \text{ MeV}$ and muon mass $m_\mu = 105.658 \text{ MeV}$ help define the maximum possible energy transfer $W_{\max}$ from a muon to an atomic electron.

Muons lose approximately 2 GeV of energy as they travel through the atmosphere from their origin, roughly 15 km above the Earth surface, to sea level, where their average energy is around 4 GeV ($\gamma \approx 38$)[9]. Consequently, many of these muons exhibit minimal interaction, falling within the broad minimum of the Bethe-Bloch curve[6].

¹From the Greek *hodos*, meaning "path" and *skopos*, meaning "observer".

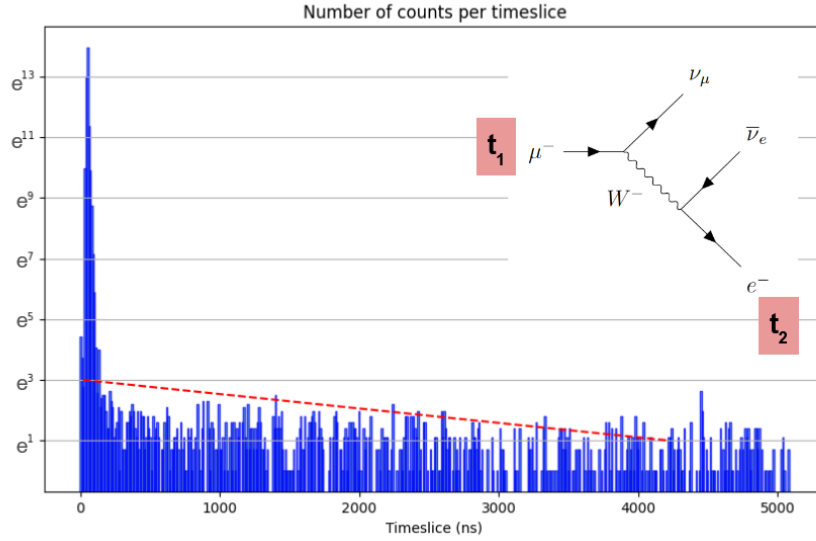


Figure 2.1: The final result of my work. The red dashed line represents the expected muon decay slope, while the blue bars show the distribution of firings in the scintillator. The distribution is generated by the initial muon passing through the hodoscope, activating the readout, and the final flash due to the electron of the decay.

The hodoscope tracks their trajectory and through the electronic chain, enables the plotting of triggered firing events that indicates the muon as it goes through the array, then possibly stops, and then decays with another flash due to the electron. This distribution of events provides a measurement of the muon lifetime.

Such a process is not feasible through simple observation and macroscopic analysis. To handle the tracking of the light produced in the scintillation phenomenon, a precise electronic chain is required to convert analog signals into digital logic, which is accomplished via a circuit board, intentionally designed for this experiment.

What I focused on in this project was the implementation of a bit-by-bit Python code data acquisition to interpret the binary files generated by the firmware and arranged in 32-bit words, where a 1 represents a hit in one specific scintillator (light detected) and a 0 indicates no hit. From this, a distribution of events over many occurrences is plotted, allowing us to extract meaningful data about the muon decay process.

Another pivotal development in the experiment was the implementation of a new automatic SiPM threshold scanning Python-based method, built to

investigate the noise and the right functioning of SiPMs. This new software significantly streamlined the process and demonstrated the experiment reproducibility, which is also fundamental for all experiments including SiPMs. Third year physics students at Cambridge University will eventually have the opportunity to observe how this prototype functions, along with the plots generated by the software, which describe the characterization of the SiPMs and scintillator firing events, that will finally reveal the expected exponential trend, allowing for the determination of the mean lifetime, τ .

2.1 The Muon Hodoscope

The prototype used for this experiment consists of an array of 25 EJ-200 organic scintillating bars made of polyvinyl toluene, arranged to form a cube with a side length of 25 cm (Figure 2.2(a)). Each scintillator is wrapped in a Tyvek layer to prevent light leakage, and it is characterized by a peak emission amplitude of 425 nm [5].

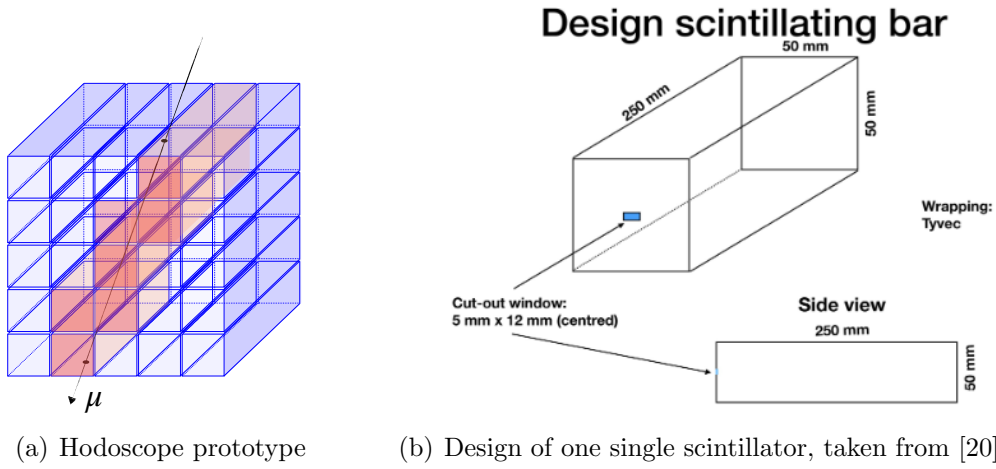


Figure 2.2: design details.

Each bar has dimensions of 50 mm $\times$ 50 mm $\times$ 250 mm and is enclosed in a specially designed brass capsule that conforms closely to the parallelepiped shape. This capsule includes an alignment with the position of the screw hole in the scintillator to secure the entire prototype correctly. Details on the assembly will be presented in Section 2.3. Preventing the influence of external light is essential for this type of experiment, as SiPMs are highly sensitive and susceptible and they are coupled to one scintillator on behalf of a tiny cut-out window as depicted in figure 2.2(b). Their functionality will be characterized, and the significant sensitivity to single photons will be evident.

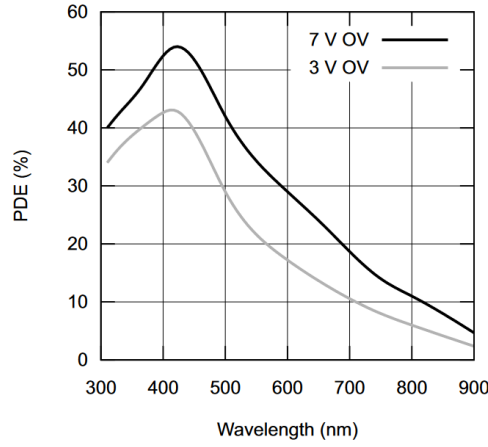


Figure 2.3: Spectral sensitivity of the SiPMs sensors used in this experiment. The peak corresponds to a waveleght of 420 nm, coherent with the peak emission amplitude of the PVT scintillators used.

2.1.1 SiPM Physics

The sensors FBR-S4N33C013 were specially realised for this experiment from a printed circuit board; they have a characteristic break-down voltage of 26.9 V and a peak efficiency PDE of 43% with an overvoltage of 3 V² [2] as shown in Figure 2.3. More generally, a SiPM is consituted by a matrix of micro-cells³. Each microcell is a photodiode operating in Geiger mode, hence an avalanche photodiode polarized at a reverse voltage greater than the break-down voltage. This creates a strong electric field in the *depleted region*, providing enough kinetic energy to a single charge carrier so that even a single electron-hole pair can trigger the impact ionization process and the avalanche multiplication of carriers.

It is clear that a photodiode in which the arrival of a photon has triggered an avalanche in Geiger mode cannot detect the arrival of a second photon. Therefore, a process is needed to stop the avalanche by lowering the electric field across the depletion region to a level where impact ionization is no longer possible.

This is achieved by using a circuit known as a *quenching circuit*. The simplest quenching circuit consists of a resistor in series with the photodiode junction.

²Overvoltage refers to the difference between the applied voltage and the breakdown voltage of the SiPM.

³These microcells are also called SPADs (Single-Photon Avalanche Diodes).

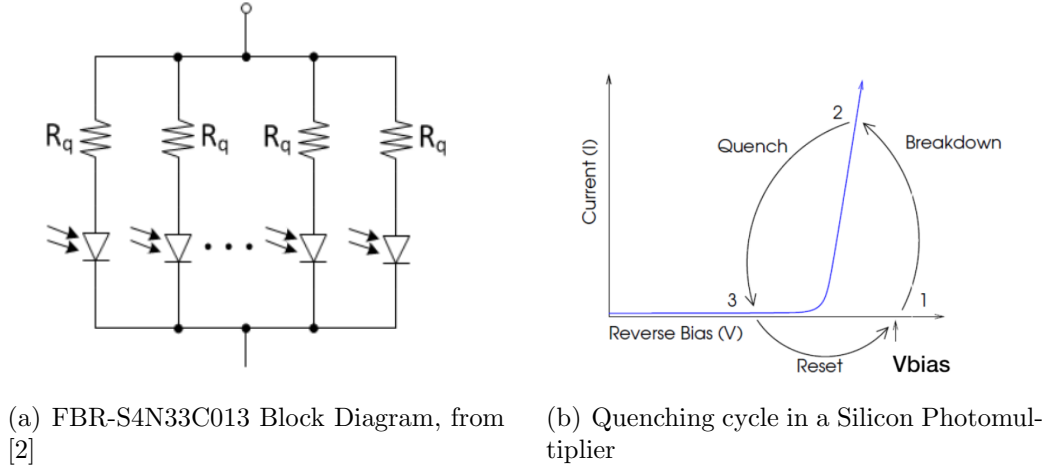


Figure 2.4: SiPMs details

In general, a SiPM must be biased with a voltage V_{bias} greater than the breakdown voltage V_{bd} , which is the characteristic voltage that needs to be applied in order to activate the avalanche. The overvoltage ranges from 2 to 4 V. V_{bd} has a value of approximately 27 V, which means that the bias voltage ranges from 29 V to 31 V.

Temperature is one of the main issues affecting the behaviour of the photodiodes inside the SiPMs and heat has an impact on gain and noise. **DCR** is the response of a SiPM giving counts even if there is no incident light hitting the device. These so-called "dark counts" mainly stem from thermal fluctuations which, like in the case of photons, can create electron-hole pairs in the semiconductor. Especially, for *single* photoelectron (p.e.) mode counts which is defined as consists of only the firing of one microcell, thermal noise is dominant source.

Optical crosstalk is a minor effect that can also fire multiple neighbouring cells, creating double or higher p.e. modes. A photon emitted in an avalanche in one cell re-triggers the neighboring cell, resulting in a signal that looks like more than one photon has been detected.

In general, a suitable threshold level is important for eliminating false signal detection from thermal noise.

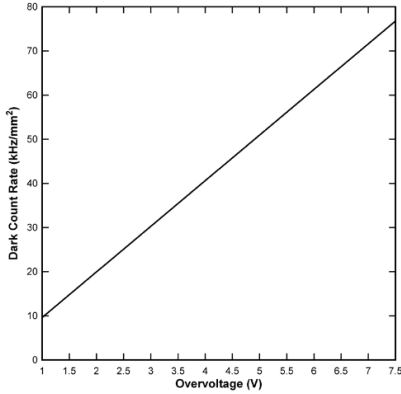
To quantify the response of the SiPM, the total charge output Q generated by the detection of multiple photoelectrons can be calculated as:

$$Q = N_{\text{fired}} \times G \times e$$

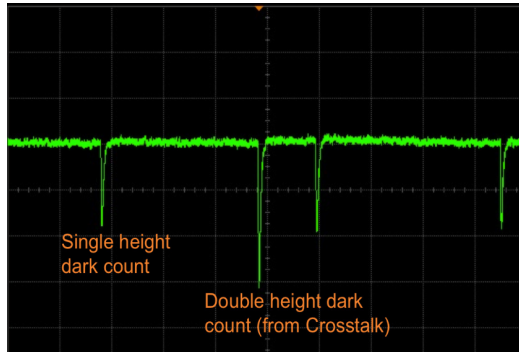
where N_{fired} is the number of microcells that fired, G is the gain of the device, and e is the elementary charge. The gain G depends on the overvoltage applied, which increases the avalanche multiplication factor (for SiPMs used in this experiment it is 3.2×10^6 [2]), amplifying the single photoelectron signal well above typical noise levels [12].

Thus, DCR is typically expressed in kHz and depends on several factors, including:

- **Temperature:** As temperature increases, the DCR increases, since thermal fluctuations become more common (doubling nearly every 8 degrees [13]).
- **Overvoltage:** Higher overvoltage (the difference between the bias voltage and the breakdown voltage) leads to a higher DCR.
- **Material quality:** Defects in the semiconductor material can increase the DCR.



(a) DCR (normalized to the surface) as a function of overvoltage [2]



(b) DCR and Optical crosstalk amplitude signals on the oscilloscope [2]

Figure 2.5: Overvoltage and DCR details.

In the threshold scan calibration procedure, DCR was deeply investigated, as well as general *electronic* noise, which was statistically coped with as a perturbation of the DCR.

2.2 From Analogic to Logic

What I have analyzed are *.dat files filled with ones and zeros, or sometimes letters in hexadecimal notation. What do these bits have in common with a spark of light produced by an ionizing muon interacting with the plastic material? There is a specific logic path with many components inside, each of them fundamental, ensuring that each of the 50 sensors (for the 25 total scintillators, with 2 sensors connected in OR logic for each scintillator) is properly read and processed.

The operation of this detection system begins when muons pass through the scintillator, which emits light in response to the interaction. This light is then detected by Silicon Photomultipliers (SiPMs), which convert the photons into analog amplified electrical signals. These signals are subsequently directed to the Power Distribution & Bias Board, which provides the necessary voltage for stable operation (30.0 V).

From there, the signals pass to the Comparator Board (printed specifically for this new prototype during my stay), which plays a crucial role in digitizing the data. Since digital electronics operate in two states (high/low, true/false, or 1/0), comparators are electronic components that literally *compare* two input signals, typically two voltages, and produce a digital output indicating which of the two signals is greater, thus converting it into a digital form. This transformation marks the point at which all further processing is digital. The organisation of how the hardware works is managed by a low level software, the firmware, which will then communicate with the final software.

The digitized signals are then sent to the *Field Programmable Gate Array* (FPGA), which organizes the 1s into specific positions within a 10 ns timeslice corresponding to a 32-bit word, a sequence that software later interprets as binary numbers. This binary information is sent to a PC and saved as a binary file. Finally, it is my turn to look into it and try to find where the τ hides.

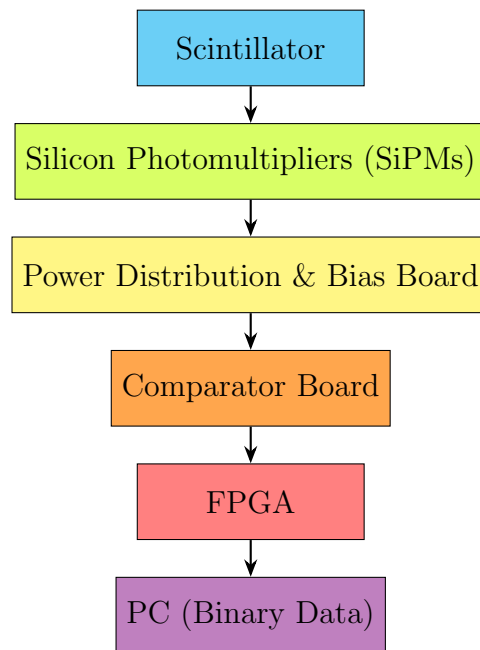


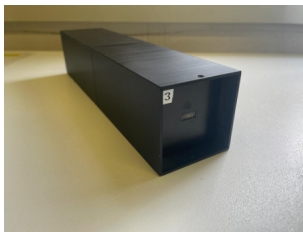
Figure 2.6: Schematic representation of the electronic chain used in the setup. The scintillator detects particles, and the signal is amplified and processed by SiPMs. The power distribution and bias board provide necessary voltages, while the comparator board refines the signal. The FPGA handles data acquisition, and the PC stores the binary data for analysis.

2.3 Mounting and Casing

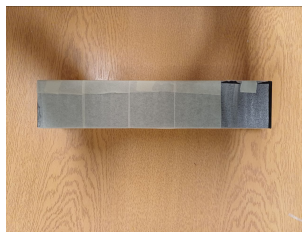
This part required high precision, primarily to achieve successful measurements, to assess this as a successful experiment, indeed. There are two main aspects we⁴ focused on:

- **light leakage** is crucial, which is why each scintillator was initially covered with a thin layer of plastic foam to create a cushion that improved adhesion to the outer brass casing. This casing is essential for preventing light leakage and for ultimately securing the entire setup physically.
- a **good contact** between the pixel matrix of the photomultiplier and the scintillator's cut-out window is fundamental for the experiment success; the use of flexible PCB⁵ SiPMs was particularly important in achieving this.

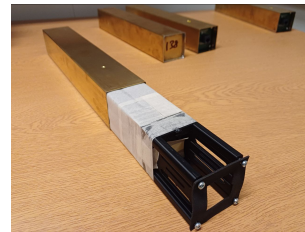
Initially, we unpacked the scintillators ordered specifically for this experiment, covered in black plastic tape to prevent light leakage. For added protection, they were wrapped with clear plastic (the same they were initially packed with) and secured with paper tape to enhance protection and adhesion within the brass case in which they will ultimately be placed.



(a) Uncovered Scintillator



(b) Covered Scintillator



(c) Brass Casing

Figure 2.7: Steps of the covering procedure for each scintillator.

As far as the contact between sensors and the cut-out window is concerned, the tolerance in the width of the vertical slots in 3D printed guides enables slight horizontal movement, allowing the PCB to align itself once the sensor is pressed into the cut-out window on top of the scintillator. The SiPM can be pushed further inward by adding an adequate number of washers between

⁴My lab mate Shikang and I collaborated for several days to mount everything.

⁵*Printed Circuit Board*. It is a board used in electronics to mechanically support and electrically connect electronic components using conductive pathways, tracks, or signal traces etched from copper sheets laminated onto a non-conductive substrate.

the vertical slot and the PCB. The PCB is then firmly held in place with screws. Figure 2.8 shows the mounting with two flexible PCBs attached. Another possibility to enhance the contact quality would be to use transparent optical grease. However, while this method could improve the optical coupling between the SiPM and the scintillator, it would risk spoiling the window and make dismounting more difficult for future adjustments or maintenance.

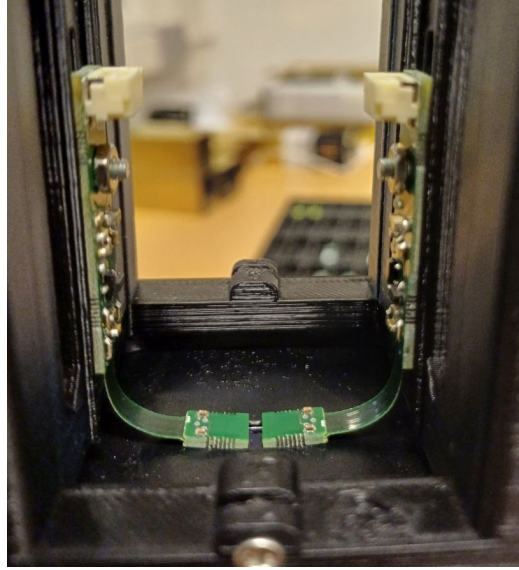


Figure 2.8: Flexible PCBs attached to the guide.

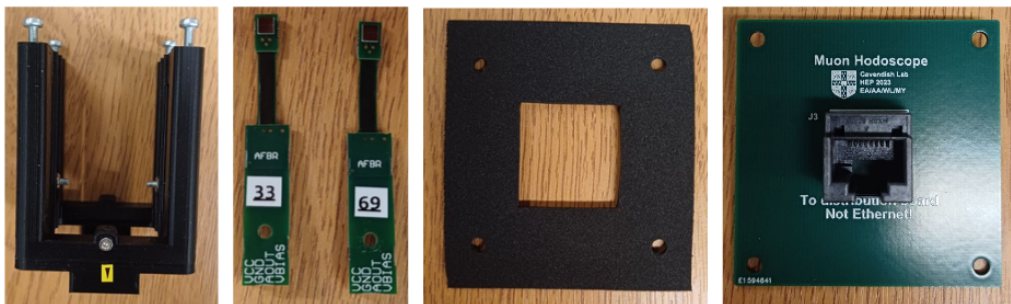


Figure 2.9: From the left: 3D printed guide to connect the scintillator to labeled photomultipliers, flexi PCB photomultipliers, foam protection to prevent light leakage, SiPM cap to cover the entire mounted assembly and connect with 4-pin wires.

White 4-pin cables were used to connect sensors to the external case creating

a correspondence between VCC labels, to finally receive the power supply from the power distribution board in the final setup. The back of each cap was previously coated with black rubber paste around the pins to prevent light leakage, as previous experiments demonstrated a significant improvement. Additionally, the initial protective films were kept in place to safeguard the cable connection area.



Figure 2.10: From the left: 4-pin cables to connect the SiPM cap and flexi PCB, ensuring alignment with the VCC side; pins coated in black rubber; assembly closed with a screwdriver to maintain a connection with the cable for board attachment; final result: upper-view of the scintillator placed within the brass casing.

	SCINTILLATOR	Left SiPM	Right SiPM
1	137	32	31
2	135	23	22
3	139	20	21
4	140	26	25
5	147	30	29
6	145	28	27

Table 2.1: Classification table after labelling every component. Here 6 out of 25 scintillators are shown as an example.

2.4 Threshold Scan

Once the apparatus is mounted and everything is connected to the power supply through the power and bias board, we check that the power supply (i.e., the bias voltage) is the same for each scintillator with a multimeter. Right afterwards, a fundamental part — and probably the core of my contribution to this experiment — is the **threshold scan**, implemented with an automated method that significantly speeds up the analysis compared to the previously used software⁶. But what is a threshold scan, and why is it useful?

Setting a threshold for each SiPM means selecting those signals whose amplitude, in voltage, is more negative (or higher in absolute value) than the triggering threshold. The threshold scan is essential for the correct calibration of the setup and ensures that the sensors are functioning in the optimal way we expect. This way, we establish the best operational settings and measure the tuning performance through a subset of scans.

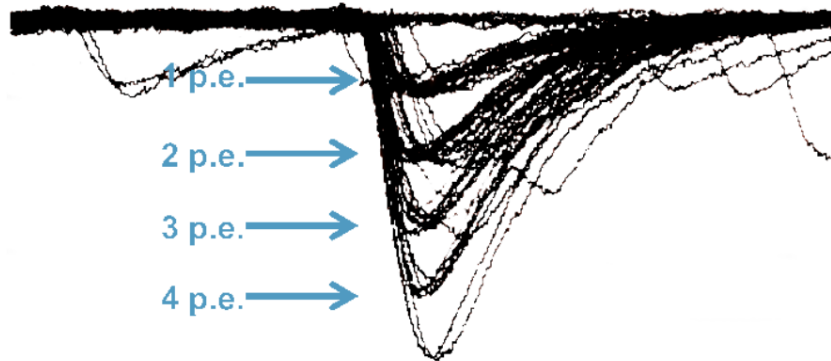


Figure 2.11: Oscilloscope shot showing the discrete nature of the SiPM output when illuminated by brief pulses of low-level light, taken from [17].

⁶My colleague Shikang had a similar experiment to mine but worked with older software, which took about 2 hours for each operation, whereas mine took just a blink of an eye.

The idea is that most of the signals selected by the photomultiplier sensor are genuine photons produced by the passage of the muon or, possibly, by the decay electron, leaving out so-called *false positives*.

Furthermore, investigating the dark count behavior of the SiPMs is important to identify the position of the different photoelectron (p.e.) modes of each SiPM with respect to the threshold voltage. As explained in Section 2.1.1 and shown in Figure 2.11, the SiPM high granularity allows a single photon to trigger an ionization avalanche and generate a measurable macroscopic current. Moreover, the SiPM can detect multiple photons at a time because of its SPAD matrix structure, consisting of multiple independent microcells. The output is discretized into multiple photoelectron modes (peaks). The

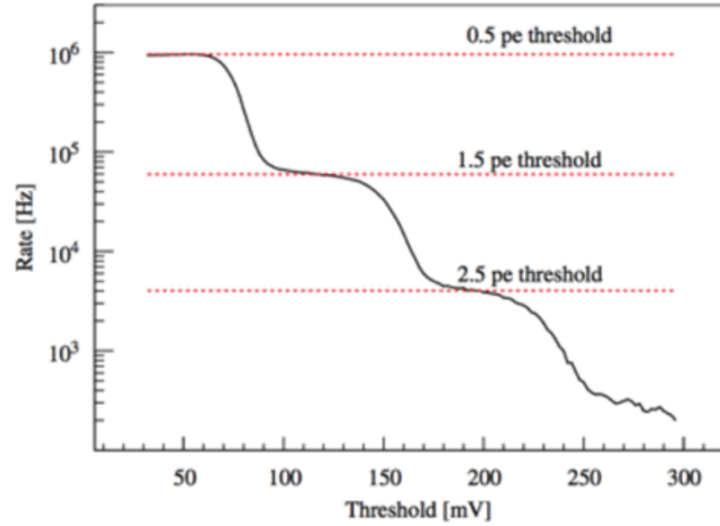


Figure 2.12: DCR frequency in Hz as a function of threshold for different biases *in absolute value*: a step trend is visible, taken from [15].

trigger voltage on the oscilloscope is set to select all signals of intersection and those below. Setting a threshold too low (i.e. , very negative) would select only real events and not noise but would miss many real signals, reducing detection efficiency. Conversely, a threshold set too high would intercept too much DCR noise. The goal of this type of calibration is to find a compromise between noise and real signals.

In this type of calibration, my role was to implement the code that enabled plotting the Dark Count Rate (DCR) (the detected frequency for each sensor) as a function of the discrete threshold voltage values. The dots are then connected into a continuous line to create the final plot for each sensor. In the

read-out, the threshold voltage is linearly transformed into a dimensionless discrete index readable by the DAC⁷, which then converts it into a physical mV. We conveniently kept units as a dimensionless index in our analysis, since converting to physical units would simply be a linear transformation of the x-axis without changing the result type. Major details on this linear transformation are thus trivial and not dealt with in this context.

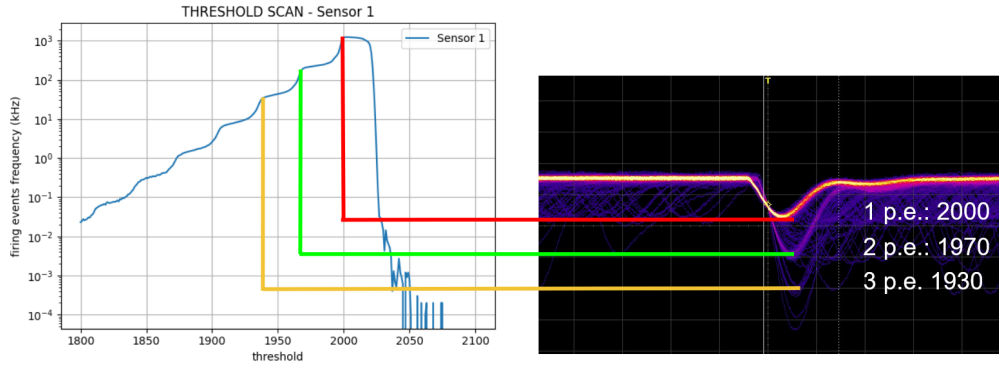


Figure 2.13: Correspondence between the photo-electron modes on an oscilloscope shot recorded in the lab and the threshold scan of the same sensor. The highest frequency peak corresponds to the first photoelectron mode.

The result is the analysis I obtained from the frequency data at various threshold values in binary format for each photomultiplier sensor, generated by the firmware logic. For those interested, I transferred my code `thresholdscan.py` to a GitHub repository, which can be found in the abstract.

Unlike in Figure 2.12, the DAC index for the threshold is higher when the set threshold is less negative (in absolute value). Conversely, a more negative threshold (i.e., a lower threshold on the oscilloscope) corresponds linearly to a lower DAC index. The staircase behavior in the plot 2.14, shows the relation between the DCR in logarithmic scale and the threshold DAC value. The frequency of counts is calculated using the formula:

$$f = \frac{N}{\Delta t}$$

where N is the number of hits returned by the data logger, and Δt represents the time interval over which the measurements were taken; in each data block, hence for each discrete threshold index, it was 10 s. Our task is to identify

⁷Digital-to-Analog Converter.

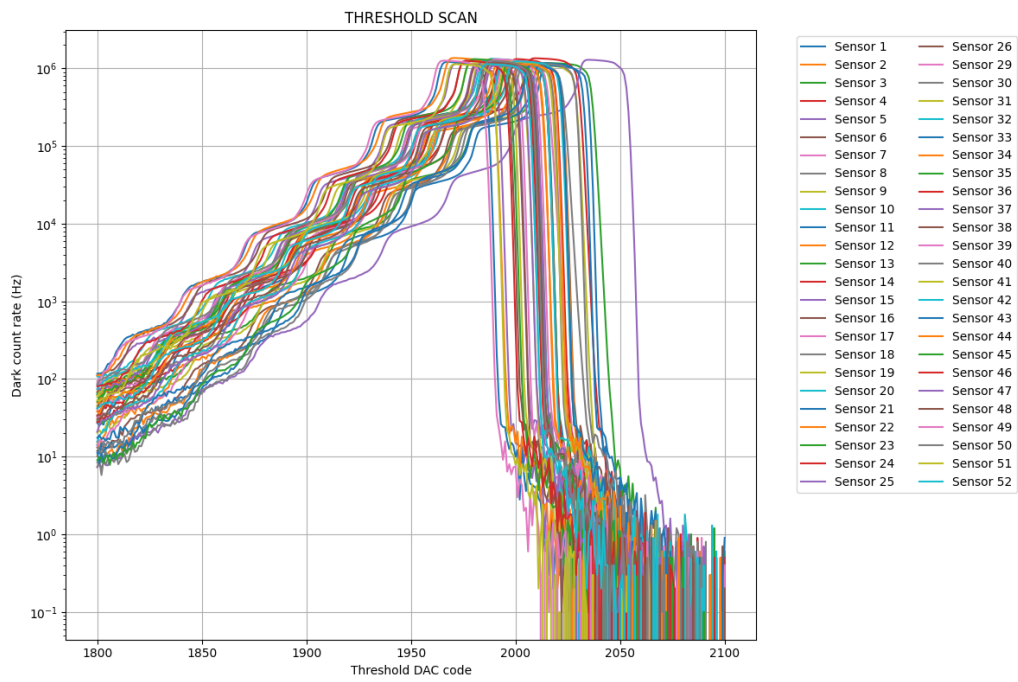


Figure 2.14: Plot generated by me, also available on the HEP Muon Hodoscope page. Apart from a horizontal shift, all sensors show the same staircase trend.

a consistent frequency across all sensors so that most signals are captured while avoiding noise signals. Initially, we noted that around the frequency of 1 kHz (visible on the y-axis in Figure 2.15), a threshold would trigger the fifth p.e. mode and higher signals. This is reasonable, considering that the most intense dark count signals visible on the oscilloscope with each sensor connection did not exceed the fifth p.e. mode.

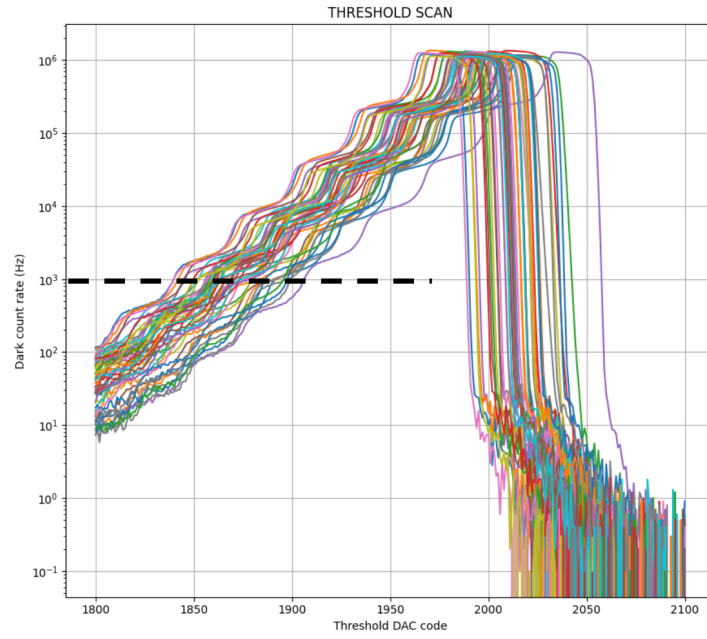


Figure 2.15: Procedure for selecting specific trigger thresholds for each sensor. Selecting a suitable counting frequency - in our case between 1 kHz and 10 kHz - likely includes real signals and limits noise within the fifth p.e. mode. The threshold scan intersects the plot at the threshold point corresponding to the fixed frequency value.

Some comments can be made on Figure 2.14. First of all, the idea of plotting all the sensors is clearly a way to show the strong consistency in the growth behavior across all 50 sensors used in the experiment. A horizontal shift can be observed, which, however, is reasonable. Sensor 25 shows the largest shift to the right, with a 70 DAC unit shift compared to the one furthest to the left (SiPM 7). This is an acceptable result, as it represents a shift of only 3-4% relative to the analyzed threshold values.

The main reasons for the observed shifts between sensors relative to the threshold are mainly due to the following factors:

1. **Sensor-to-Sensor Variability:** It is intrinsically introduced in every sensor by manufacturing. These differences bring about small discrepancies in sensitivity and detection efficiency, thus resulting in the sensors performing differently for the same threshold level. This intrinsic variability underlines how a threshold scan is an indispensable calibration for every sensor, where a proper threshold should be set.
2. **Noise levels and background fluctuations:** Each sensor has a unique noise profile that affects the threshold level above which a given sensor initiates a count detection. Changes in background noise are responsible for these differences, hence even sensors under the same conditions exhibit shifts from one to another.
3. **DAC Characteristics:** Sometimes, the behaviour of the DAC itself may bring in minor inconsistencies; hence, the data capture has some nonlinear behaviour.

As what concerns the third reason, we conducted tests by powering the system off and on, as shown in Figure 2.16, and the results were reproducible across multiple runs. This consistency suggests that the DAC is stable and not contributing to the observed variability in this setup. Since this experiment is intended for use by third-year physics students, reproducibility is essential in experimental physics, and it's also helpful for them as it keeps the conditions consistent from one day to the next. Another type of noise is the electronic one, which will be particularly dealt with in Chapter 3, with a deep statistical perspective.

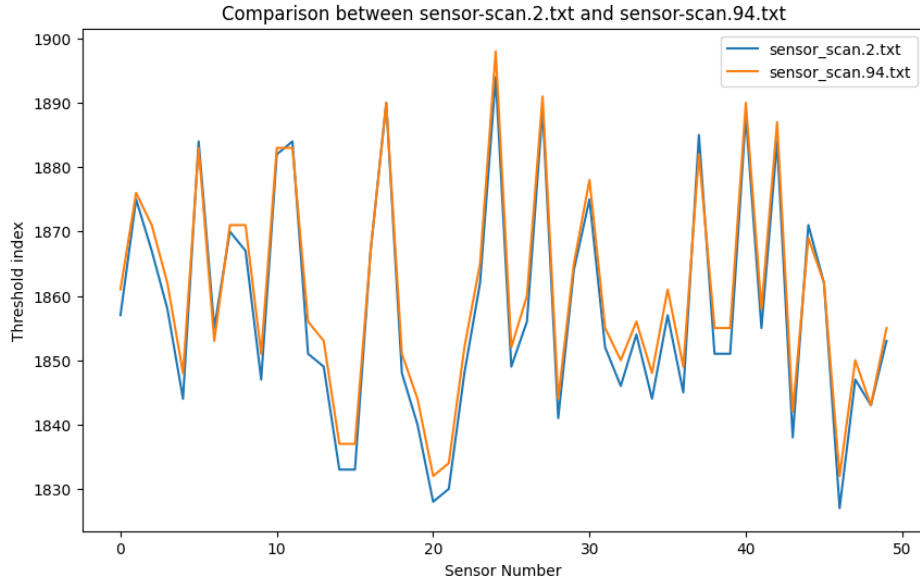


Figure 2.16: Under the same conditions, threshold index values for two different data acquisitions, separated by the shutdown of the entire power supply. The two scatter plots, made continuous by a broken line, show strong consistency under identical conditions, despite the intermediate shutdown.

This new automated scan provided several advantages:

- It allows us to quickly find the optimal working point for a large number of sensors simultaneously.
- It enables rapid identification of faulty sensor channels.
- It is an essential step before data acquisition to ensure a uniform detector response to muons across the 25 scintillators.

In conclusion, initializing an automated threshold scan allows for the evaluation of a noise frequency data acquisition as a function of a threshold index and streamlines the result into a text file that contains the optimal threshold for each sensor, based on a fixed frequency deemed appropriate after observing the photoelectron modes on the oscilloscope.

2.5 Firing Events

What we want to eventually reveal is the path of cosmic ray muons (naturally occurring charged particles travelling at speeds close to the speed of light) as a pattern of light produced as the muon traverses the stack of bars.

How can then 0s and 1s give us information about the muon trajectory and, eventually, its mean lifetime?

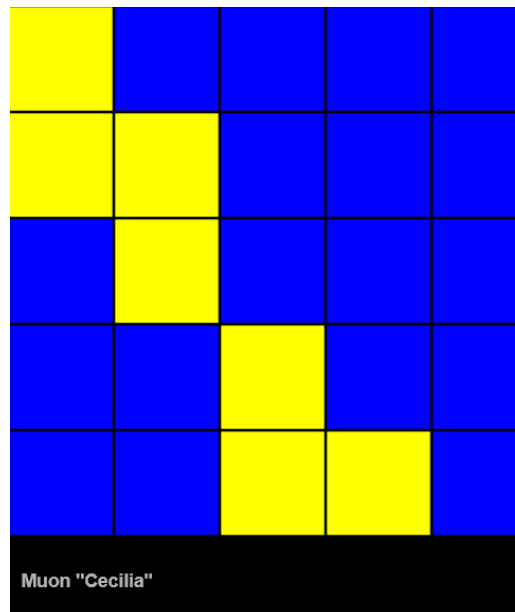


Figure 2.17: Muon "Cecilia" GUI trajectory, one of the first revealed. Yellow squares indicate the firing scintillators, hence where the muon passes. A GIF slideshow of muons can be seen on the Muon Hodoscope HEP page, linked in the abstract.

2.5.1 Data Acquisition Communication Protocol for Muon Detection

The purpose is to record muon events and potential decay signals, particularly identifying cases where muons stop within a scintillator array prototype, allowing for analysis of initial muon signals and any secondary signals from decay processes.

The experiment setup and program are designed with reproducibility in mind, as proven in section 2.4 . Consistency in acquisition conditions ensures reliable analysis across sessions, making results comparable from one acquisition

period to another. Each data acquisition event follows a defined structure, coming from the firmware logic settings, to ensure uniform data collection. The data organization includes the following elements:

- **Header and Final Header:** each packet of data is framed with a 32-bit word header at the start, `0x030201CB`, followed by 3 words due to the reading delay, and a final header, `0xFFFFFFFF` at the end, marking the packet boundaries for segmentation and organization.
- **512-Word Format:** beyond the initial 4 words, data packets are in format of 512 words, namely 2^9 . Each word corresponds to a timeslice of 10 ns. This yields a consistent length to analyze the muon mean lifetime⁸ and can be streamlined for processing and analysis.
- **Muon Triggering Mechanism:** The readout of the system is triggered when a muon passes through the scintillator. More precisely, firmware logic is implemented to tag it as a muon with an A&&B&&C logic, meaning that a flash is recorded whenever at least the first top three layers of the array are fired in squares that are connected either by a side or a common vertex. This ensures that only those signals that act as an indication of muon presence are recorded and unrelated noise is sifted at the first instance, hence increasing accuracy in the data.

In this setup, one is interested in the relative timing between signals rather than absolute time of detection. The goal will be finding the time difference between initial muon detection and any secondary signal that would indicate electron decay from a muon that has stopped within the scintillator array. This approach will be very important in studying the physics of the decay events in great detail.

⁸Nearly 5 μ s, reasonable time window to analyze τ .

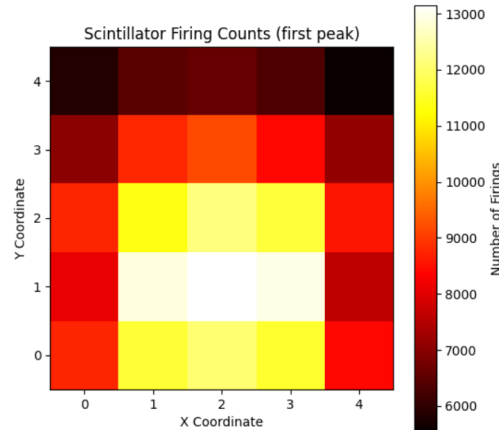


Figure 2.18: Heatmap plotting over multiple muon events, demonstrating uniformity. In reality, however, the coordinates are inverted, showing the opposite orientation.

2.5.2 Program Functionality and Structure

The code I initialized provides valuable information about the spatial positioning through the activation of specific scintillator bars. By mapping each scintillator based on the number of firings, a heatmap of events can be created.

It hence allows to trace the muon's trajectory by analyzing the initial triggering words, as shown in Figure 2.17. Alternatively, by examining a larger dataset, we can observe a uniform distribution of firings (Figure 2.18). This procedure has helped identify some issues caused by abnormal retriggering, which will be discussed in Section 2.6.

- Bit Identification and Scintillator Mapping:** The program is designed to identify each bit that corresponds to a specific scintillator. In the initial tests, each scintillator was connected individually to determine which bit in the data frame was activated. This setup ensures accurate tracing of detected signals back to their respective scintillators.
- Scintillator Class Structure:** Each scintillator is represented by a dedicated class in the unpacking code. This structure allows for modular data handling, and the program unpacks binary data based on the specified header and final header. The unpacking process is crucial for correctly associating each signal with the corresponding scintillator, as illustrated in the code snippet below.

- **32-Bit Data Frame Analysis:** The data frame consists of 32 bits, with specific bits assigned to each scintillator. When a muon triggers a scintillator, the corresponding bit is activated, capturing data for that particular scintillator. If there is no activation, the bit remains inactive. This selective analysis reduces the processing load, minimizes false positives, and improves data accuracy.

```
# Class representing a scintillator, including its ID, grid
# position (x, y),
# physical position (phys_x, phys_y), and the timestamp.
class Scintillator:
    def __init__(self, id, x, y, phys_x, phys_y, t):
        self.id = id
        self.x = x
        self.y = y
        self.phys_x = phys_x
        self.phys_y = phys_y
        self.time = t

    # String representation for better readability when
    # printing scintillators
    def __repr__(self):
        return f"Scintillator(id={self.id}, x={self.x}, y={
            self.y}, phys_x={self.phys_x}, phys_y={self.phys_y
            }, time={self.time})"

# ...

# Function to decode a single 32-bit word into active
# scintillators at a given time
def decode_word(word_int, time):
    active_scints = []

    # Process active bits 0-12, then skip 13 and 14...

    # Process active bits 15-26 (IDs 14-25):
    scint_id = i
    if word_int & (1 << i): # If the bit is set, the
        # scintillator is active
        x = (scint_id - 1) % 5 # Grid x-coordinate
        y = (scint_id - 1) // 5 # Grid y-coordinate
        phys_x = x * 5 + 2.5 # Physical x-coordinate
        phys_y = y * 5 + 2.5 # Physical y-coordinate
        active_scints.append(Scintillator(scint_id, x, y,
            phys_x, phys_y, time))

    return active_scints
```

2.5.3 Visualization and Data Interpretation

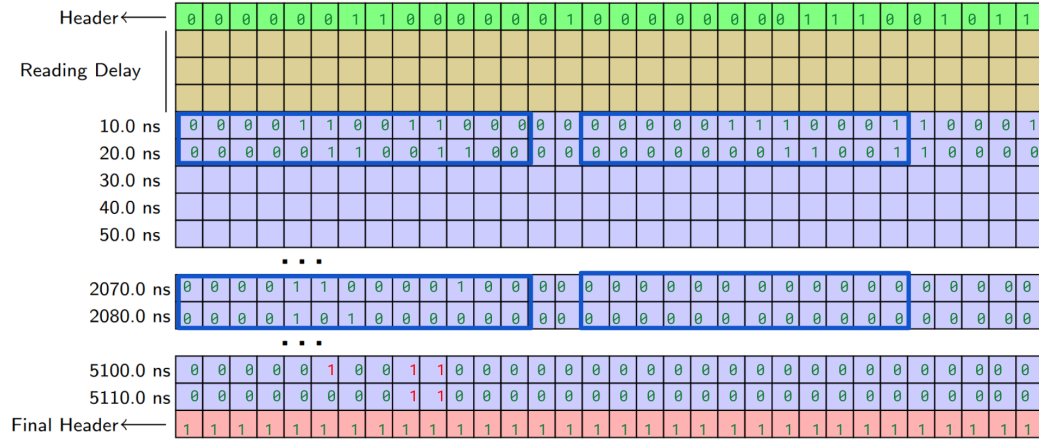


Figure 2.19: The visualization displays each bit in the data frame, mapping each bit to a specific scintillator (contoured in blue). This graphical layout helps in understanding the link between bit activations and scintillator positions. Red 1s are an example of noise, which plays a crucial role in this type of experiment, as well as on the debugging part.

Visualization is a key part of this protocol, particularly for interpreting muon decay data distribution. A Beamer-format visual representation (Figure 2.19) is created to illustrate the bit structure. The plot presented at the beginning of the chapter, initially just a preview, is now fully understandable. The idea is to plot the decay as a distribution over many detected events, a percentage of which — those with lower energy — stop within the array and, once trapped, undergo their decay, generating a second flash. The bit structure selectively highlights bits outlined in blue, totaling 25 out of 32. These include the first 13 bits, skip two disconnected ones, and then the remaining 12 bits.

2.6 Debugging

In physics, one must remember that results do not just appear instantly. It takes patience, perseverance, and sometimes a little humor to accept that things do not always work as we want, especially the first time. During our data analysis, some anomalous phenomena arose — a subtle reminder that every experiment has its hurdles, no matter how perfectly designed.

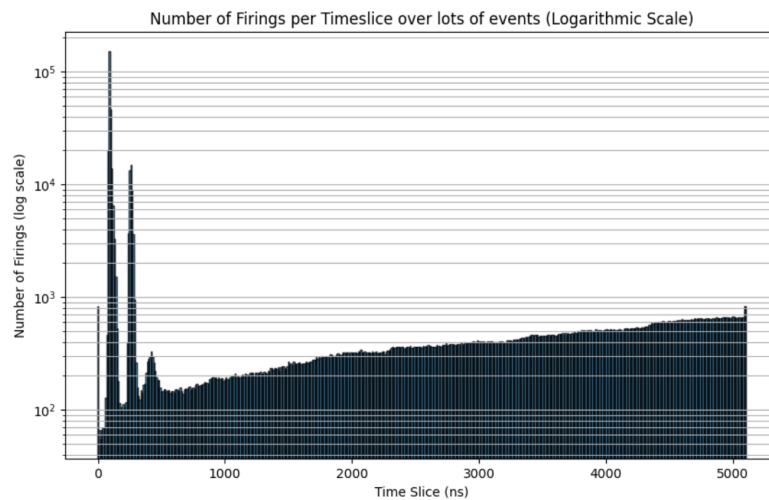


Figure 2.20: Not exactly a distribution that resembles a muon decay.

2.6.1 Analysis of Anomalous Retriggering Behavior in Firmware Logic

This behaviour was observed through meticulous, word-by-word analysis of the data events, where we encountered a small percentage of cases lacking the typical final header `0xFFFFF`. Instead of cleanly ending as expected, these cases triggered a warning message due to an unexpected and persistent noise pattern.

In certain cases, as we parsed through the binary file, we stumbled upon a sequence resembling:

```
50000000  60000000  00000000  70707070...
```

Upon encountering this specific pattern, the firmware logic began to behave abnormally, generating a repeated `70707070` sequence in what seemed like an endless loop. This suggests that the trigger logic in the firmware misinterpreted this sequence as a persistent signal, which led to anomalous

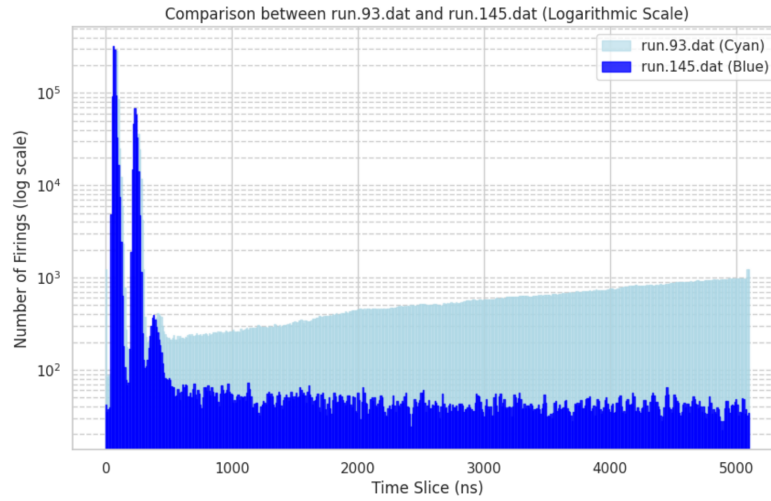


Figure 2.21: After fixing the firmware logic.

retriggering and a noisy end to the file.

After identifying this issue, we investigated the firmware logic and implemented a fix to handle these sequences correctly. By adjusting the interpretation mechanism, the firmware can now recognize this pattern and avoid entering a retriggering loop. Thus, the anomalous noise at the end of the file has been resolved, and the system now concludes its output without generating unnecessary warnings or repetitive noise.

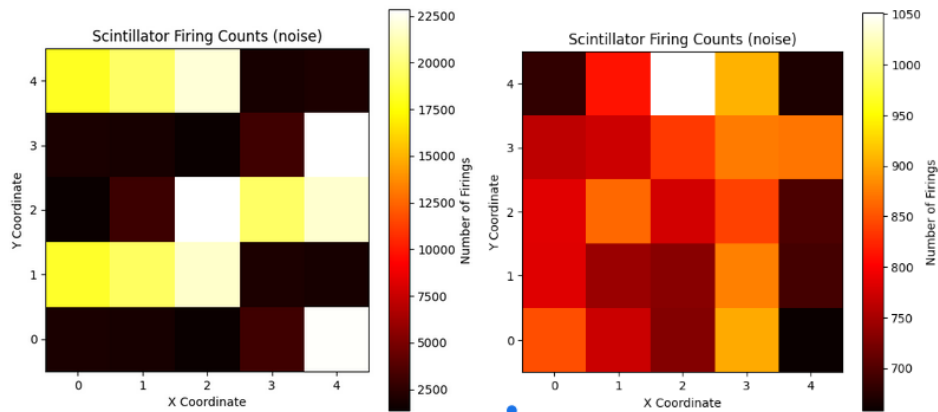


Figure 2.22: Heatmap of firing events in each scintillator in the *noise* time area, before and after fixing. Notice how more homogeneous is the scale in the second heatmap.

2.6.2 Analysis of Periodic Retriggering in Trigger Logic

By zooming in on the histogram of the number of firings per timeslice, another issue became apparent: instead of a single trigger, three anomalous peaks were observed, indicating periodic retriggering. This periodicity of 16 timeslices, occurring with a frequency that suggests a power-of-2 relationship, points to a potential retriggering problem within the electronics.

To better understand the origin of these peaks, we investigated the electronic trigger mechanism. It turns out that these retriggered peaks are caused by the comparator retriggering on the trailing edge of the signal, which is approximately 160 ns wide. This retriggering occurs due to a delay in the reset or release process of the comparator, causing multiple firings for what should be a single event. To mitigate this issue, a preliminary filtering so-

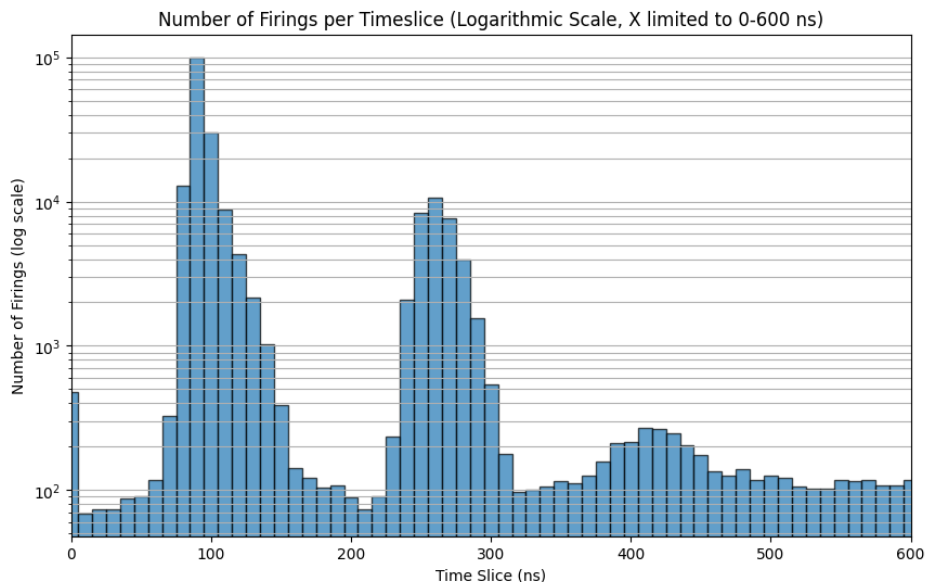


Figure 2.23: Zoom on the three anomalous peaks, separated by about 16 timeslices each, which may be possibly related to a periodic retriggering issue.

lution was applied: any hits from an event are ignored if they have already been recorded earlier in the same event. This quick fix effectively reduces the impact of retriggering, though it may introduce a slight bias, particularly in bins with lower lifetimes. However, the impact appears minimal, and the bias is not overly dramatic.

In addition to the retrigger filter, an extra requirement was introduced to improve data clarity: a time slice in an event must contain *at least two hits*

to be included in the plot. This effectively reduces background noise by excluding isolated or spurious hits, while retaining the structure of the relevant events. Although this requirement cuts down on noise significantly, it also helps maintain the fidelity of the signal, allowing for a clearer analysis of muon trajectories and decay events. Hence, what comes out of the oven is the plot you see at the beginning of the chapter. We are there!

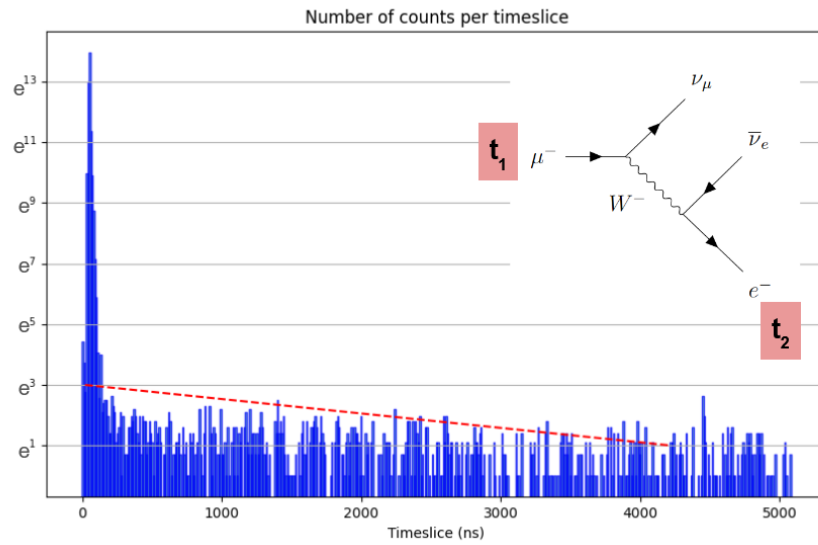


Figure 2.24: The dark side of the muon :) .

Chapter 3

Error Analysis

The aim of this chapter is to deeply deal with the physical uncertainties of what we have done, a fundamental procedure to quantify the electronic noise in the acquired data and to understand the physics behind our observations.

3.1 Threshold Scan Analysis

In the threshold scan for Silicon Photomultipliers, one can focus on the transitions between the $(n + 1)^{\text{th}}$ and the n^{th} photoelectron mode, namely the vertical part of the staircase plot. The analysis of the threshold scan involves understanding both the horizontal and vertical parts of the step curve and their respective sources of error.

3.1.1 Horizontal Part

The *horizontal* part of the threshold curve corresponds to counting the number of random events over a fixed time interval. The error associated with this counting follows a Poisson distribution and can be approximated by:

$$\sigma_{\text{horizontal}} = \sqrt{N}, \quad (3.1)$$

where N is the total number of counted events. This type of error is typically small, as it represents the statistical fluctuation inherent in the counting process. Involving a great number of events and selecting a reference frequency of at least 1 kHz, which is the result of a counter of events over 10 seconds per each threshold index. We thus keep studying at least $10^3 \text{ Hz} \times 10 = 10^4$ events per threshold index. This means that

$$\frac{\sigma_{\text{horizontal}}}{N} \leq \frac{\sqrt{10^4}}{10^4} = 1\%$$

which is reasonably low. A slight modification of this error can be done, considering the timestamp error, which is anyway far lower — 0.01%, determined by a crystal oscillator.

3.1.2 Vertical Part

The vertical part of the curve, where the threshold crosses the peak of the analogue signal, is particularly important due to its sensitivity to electronic noise. The slope in this region indicates how rapidly the transition occurs as the threshold moves past discrete photoelectron levels.

From our observations, lower p.e. modes (e.g., transitions corresponding to the 1st p.e. mode) exhibited sharper transitions compared to higher p.e. modes (e.g., 4th p.e. mode).

The signals associated with lower p.e. modes are in fact more distinct as they represent events with fewer charge fluctuations. As a matter of fact, igher threshold values tend to select events where the SiPM gain is more uniform, thus reducing the effect of electronic noise and gain fluctuations[4].

For lower p.e. modes, the number of released photoelectrons is smaller, which can result in a better-defined Signal-Noise Ratio (SNR). As the number of p.e. increases, the cumulative variation in the gain of individual microcells leads to greater overall response dispersion, hence the readout electronics might introduce more significant noise contributions[16].

Another factor can be attributed to a larger temporal dispersion (*jitter*) across microcells. This temporal dispersion contributes to the broadening of transitions at higher p.e. levels [18].

Gaussian Approximation and Noise Estimation

To quantify the dispersion of these transitions, the slope of the vertical section in the threshold scans was analyzed for one sensor, which is a reasonable procedure given the uniformity of all the sensors except for the horizontal shift, which does not affect the overall trend. Using a linear regression on logarithmically scaled frequency data (fig. 3.1), the slope (m) was computed for each transition. This parameter helps estimate the uncertainty in determining the mean threshold (μ) and the standard deviation (σ) of the Gaussian approximation.

$$\sigma_{\text{vertical}} = \frac{1}{|m| \cdot 2.355}, \quad (3.2)$$

where the factor 2.355 converts the standard deviation to the full width at half maximum (FWHM) of the Gaussian distribution. This estimation reflects the amount of noise present in the signal and its impact on the sharpness of the transition, which should be ideally vertical, with an infinite slope, and no width: a Dirac delta, indeed.

An alternative could have been measuring the number of threshold indexes comprised between 10% to 90% of the transition maximum value. While this approach is less direct in correlating with Gaussian noise, it provides a pragmatic estimation of the noise affecting the sensor performance.

Transition	Mean (μ)	Std Dev (σ)
5	1870	7.38
4	1900	7.23
3	1930	6.61
2	1960	5.41
1	1990	4.42

Table 3.1: Mean and Standard Deviation for Selected 5 Transitions sensor 1

The analysis hence shows that the SiPM operates correctly, especially at higher thresholds, where the transitions are more defined. The wider deviations at lower thresholds are consistent with expectations, given the characteristics of the electronic noise and the intrinsic noise of the device.

If the distribution narrows at higher thresholds and shows more vertical transitions, this is a positive sign that the SiPM is effectively distinguishing relevant signals from electronic noise.

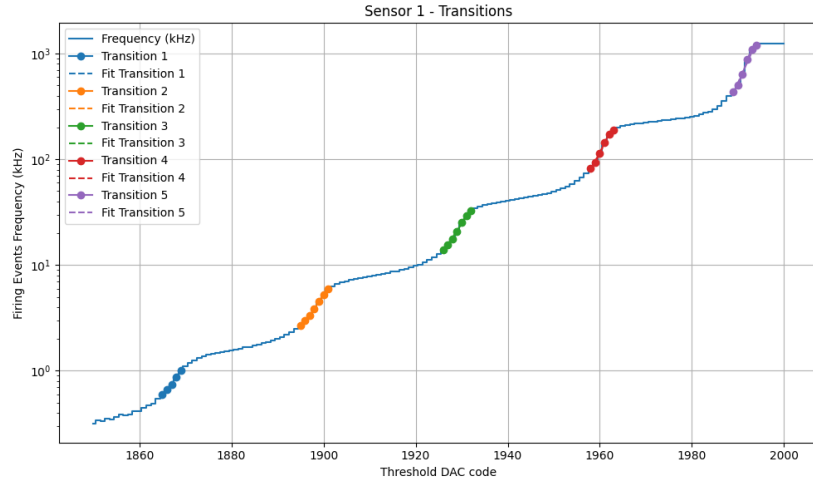


Figure 3.1: Logarithmic frequency plot with linear fits for selected transitions.

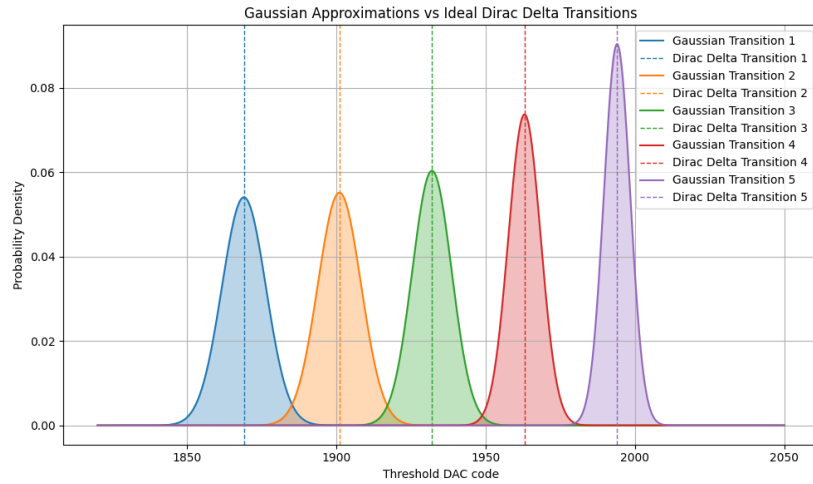


Figure 3.2: Gaussian approximations of transition slopes compared to ideal Dirac delta (vertical) transitions.

3.2 Timing Error in Firing Events Analysis

The precision of the timestamp is particularly relevant in bit-by-bit analysis of firing events, which is determined by a crystal oscillator with an estimated accuracy of 0.01%. This makes the timing error insignificant compared to the 10 ns discrete sampling, which can in contrast introduce an uncertainty

estimated as:

$$\sigma_{\text{timestamp}} = \frac{10 \text{ ns}}{\sqrt{12}}, \quad (3.3)$$

where $\frac{1}{\sqrt{12}}$ is the RMS¹ value of a normalized uniform distribution. This error contributes to the overall precision of the timing in both the detection of muons and subsequent decay events, hence affecting the accuracy in the recording of the event sequence and timing resolution.

The measurement time interval for counting dark counts, as mentioned before, is instead ideally fixed. However, in practice, minor variations may arise due to the non-deterministic behavior of computing systems. While generally negligible, this variability should be recognized as part of a comprehensive error analysis.

¹Root mean square.

Conclusions

Theory and experiment in physics are not isolated compartments; they are two perspectives of the same grand endeavour. For this reason, I have aimed to provide a comprehensive view that encompasses both perspectives, each with distinct but equally enriching methods. Muon decay serves as a model that, in theory, marks the beginning of the study of weak decay, parity violation, and interaction with matter — fundamental aspects of particle physics. The most interesting considerations to end this thesis come from the experimental perspective, though, with my direct involvement in the work.

Polyvinyltoluene (PVT) is a very nice solution for detecting charged particles like muons due to its balance of efficiency and durability. When traversing the material, a muon primarily loses energy through ionization and excitation, as described by the Bethe-Bloch formula. For a muon with kinetic energy around 4 GeV, classified as a minimally-ionizing particle (MIP), the energy loss in five layers of EJ-200 scintillators is estimated to be on the order of $\mathcal{O}(10)$ MeV. This means that even at high energies, the muon's stopping power in PVT remains relatively low [12]. In practice, only a small fraction of lower-energy muons will be fully stopped in the scintillators, which limits our ability to observe the complete decay process. Augmenting the stopping probability is therefore a pivotal point for future investigation. One possible improvement involves adding 2 cm lead plates to the setup. Lead would effectively decelerate muons, increasing the probability of capturing lower-energy muons in the scintillator array. Nevertheless, a more practical alternative — aluminum — will be more likely used due to its cost-effectiveness and non-toxic features.

As far as the code implementation is concerned, the software is essential for structured data acquisition and analysis in muon detection studies. By accurately mapping each bit to a scintillator, this new program enables precise monitoring and verification of signal paths and trigger events. What was found is that the work can be carried out with the same options and condi-

tions, allowing for reproducibility. This is essential in experimental physics, especially in educational settings, as it allows students to observe consistent results across trials.

With a Python based architecture, the current code is sometimes heavy. Transferring it to C++, a lower-level programming language, could be a sensible development to accelerate the process, especially if we aim to conduct long data recordings to achieve better statistics on muon distributions.

A more refined filtering method could also be implemented, where the affected channel is excluded only during the retriggering period rather than for the entire event window. While these drastic solutions address the current issues, ongoing refinement of the filter parameters may further optimize data quality, ensuring reliable results for muon analysis.

Finally, when addressing the possible uncertainties and errors due to the various analyses and the functioning of the electronics, a type of Gaussian broadening was observed regarding the analyses in the threshold transitions. Potential errors in the bit-by-bit data acquisition were also assessed, which are, in any case, negligible and do not affect the final distribution result, except in the case of the time window with an error from a uniform distribution for cosmic rays.

The new protocol provides a robust framework for data acquisition, facilitating efficient and accurate analysis for muon decay experiments, thereby supporting both research and educational objectives.

Returning to the initial plot, while it may still seem somewhat preliminary, this experiment offers an exciting and educational experience for undergraduates. The setup and analysis process have significant educational value, providing hands-on insight into particle physics.

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